

Determining reaction mechanisms

39

Connections

➡ Building on

- Mainly builds on [ch12](#)
- Acidity and basicity [ch8](#)
- Carbonyl reactions [ch6](#), [ch10](#), & [ch11](#)
- Nucleophilic substitution at saturated carbon [ch15](#)
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- Electrophilic and nucleophilic aromatic substitution [ch21](#) & [ch22](#)
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Arriving at

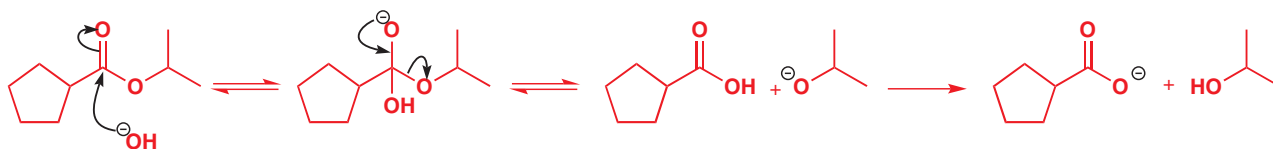
- Classes and types of mechanisms
- Importance of proposing a mechanism
- Structure of the product is all-important
- Labelling and double labelling
- Systematic structure variation and electronic demand
- The Hammett relationship explained
- Deuterium isotope effect (kinetic and solvent)
- Specific acid and specific base catalysis
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➡ Looking forward to

- Asymmetric synthesis [ch41](#)
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There are mechanisms and there are mechanisms

There are two types of answer to the question ‘What is the mechanism of this reaction?’. If you were asked to draw the mechanism of an ester hydrolysis in basic solution you should have no trouble in giving a good answer of the first type. It wouldn’t matter if you had never seen this particular ester before or even if you knew that it had never actually been made, because you would recognize that the reaction belonged to a class of well-known reactions (carbonyl substitution reactions, Chapter 10) and you would assume that the mechanism was the same as that for other ester hydrolyses. And you would be right—nucleophilic attack on the carbonyl group to form a tetrahedral intermediate is followed by loss of the alkoxide leaving group and formation of the anion of the carboxylic acid.

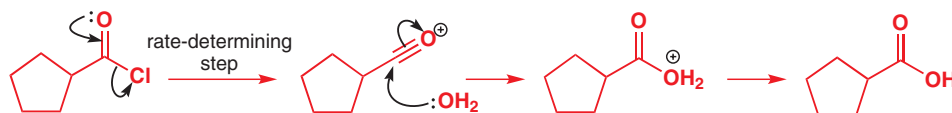


Yet someone at some time had to determine this mechanism in full detail. That work was done in the 1940s to 1960s and it was done so well that nobody seriously challenges it.

Online support. The icon  in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1**, **123-2** etc. (replacing **123** with the page number) for access to successive links.

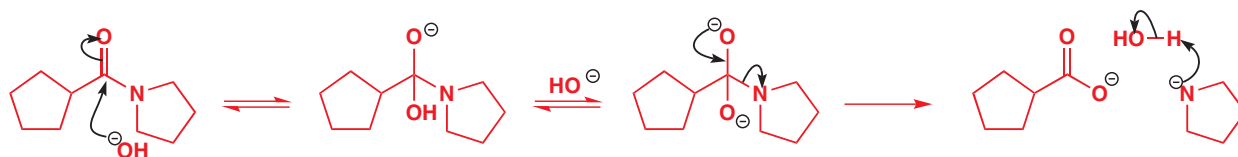
➡ This mechanism is described on p. 259.

You might also recall from Chapter 12 that, if we change the carbonyl compound to an acid chloride, the mechanism may change to an S_N1 reaction with an acylium ion intermediate because the leaving group is now much better: Cl^- is more stable (less basic) than RO^- . It would not be worth using hydroxide for this reaction: as the first step is the slow step, water will do just as well. Again someone had to determine this mechanism, had to show which was the slow step, and had to show that leaving group ability depended on the $\text{p}K_a$ of its conjugate base.



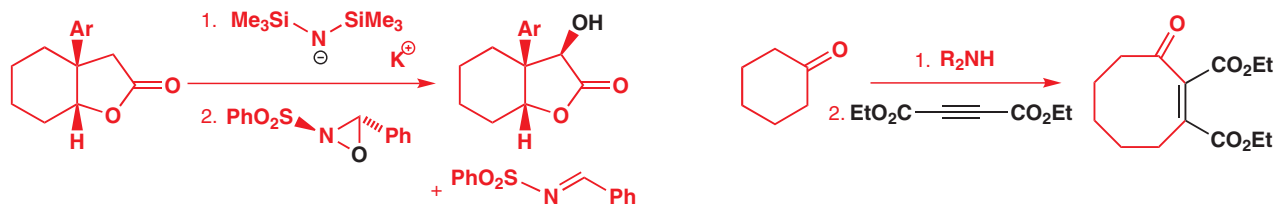
➡ Third-order kinetics, and this amide hydrolysis mechanism, were discussed on p. 260.

If the reaction were the hydrolysis of an amide, you might remember from Chapter 12 that third-order kinetics are often observed for the expulsion of such bad leaving groups and that this extra catalysis makes it worthwhile using concentrated base. Again, someone had to find out that: (1) the slow step is now the decomposition of the tetrahedral intermediate, (2) there are third-order kinetics involving two molecules of hydroxide, and (3) the first molecule acts as a nucleophile and the second as a base.



These three mechanisms are all versions of the same reaction. For you, writing these mechanisms chiefly means recognizing the type of reaction (nucleophilic substitution at the carbonyl group) and evaluating how good the leaving group is. For the original chemists, determining these reaction mechanisms meant: (1) determining exactly what the product is (that may sound silly, but it is a serious point), (2) discovering how many steps there are and the structures of the intermediates, (3) finding out which is the slow (rate-determining) step, and (4) finding any catalysis. This chapter describes the methods used in this kind of work—the detailed, second type of answer to ‘What is the mechanism of this reaction?’.

Now, suppose you were asked what the mechanisms of the next two reactions might be. This is a rather different sort of problem as you may well not recognize any of these reagents and you probably cannot fit any of the reactions into one of the classes you have seen so far. You may not even see at once which of the three main classes of mechanism you should use: ionic, pericyclic, or radical.



You may do your best to write a mechanism based on your understanding of organic chemistry, moving the electrons from nucleophiles to electrophiles, choosing sensible intermediates, and arriving at the right products. You would not claim any authority for the result, but you would hope, as an organic chemist, to propose one or more reasonable possibilities.

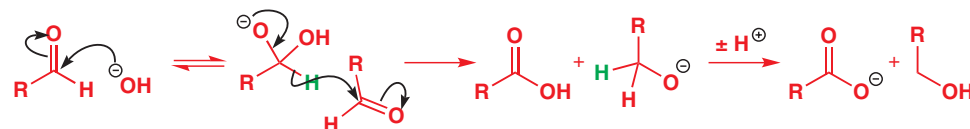
This process of proposing reasonable mechanisms is actually an essential preliminary to answering the question in the second way—finding the real, experimentally verified, mechanism for the reaction. We will now look at some of the techniques used to find such answers with an old curiosity of a reaction, the Cannizzaro reaction, as an example.

Determining reaction mechanisms: the Cannizzaro reaction

So how do we know the mechanism of a reaction? The simple answer is that we don't for certain. Organic chemists have to face situations where the structure of a compound is initially thought to be one thing but later corrected to be something different. The same is true of mechanisms. It is the nature of science that all we can do is try to account for observations by proposing a hypothesis. We then test the hypothesis by experiment and, when the experiment does not fit the hypothesis, we must start again with a new hypothesis. This is exactly the case with mechanisms. When a new reaction is discovered, one or more mechanisms are proposed; evidence is then sought for and against these mechanisms until one emerges as the best choice. That one then remains the accepted mechanism for the reaction until fresh evidence comes along that does not fit the mechanism.

We are going to look at one reaction, the Cannizzaro reaction, and use this to introduce the different techniques used in elucidating mechanisms so that you will be able to appreciate the different information each experiment brings to light and how all the pieces fit together to leave us with a probable mechanism. Under strongly basic conditions, an aldehyde with no α hydrogens undergoes disproportionation to give half alcohol and half carboxylate. Disproportionation means one half of the sample is oxidized by the other half, which is itself reduced. In this case, half the aldehyde reduces the other half to the primary alcohol and in the process is oxidized to the carboxylic acid. Before the discovery of LiAlH_4 in 1946, this was one of the few reliable ways to reduce aldehydes and so was of some use in synthesis.

Here is a simple mechanistic scheme of what happens—the sort of thing you might reasonably propose if you had not seen the reaction before.



It's not the only possible mechanism by any means—and you may spot that it is slightly different from the one in Chapter 26, where we showed a dianion as an intermediate. We'll now work through some of the alternative mechanisms that have been proposed for the Cannizzaro reaction, along with the evidence for or against them. Most of these alternatives have been eliminated, leaving just the ones you have already met. Finally, we will see that even these mechanisms do not explain everything absolutely.

Proposed mechanism A: a radical mechanism

Early on it was thought that the hydrogen transfer might be taking place via a radical chain reaction. If this were the case, then the reaction should go faster if radical initiators are added and it should slow down when radical inhibitors are added. When this was tried, there was no change in the rate, so this proposed mechanism was ruled out.

Kinetic evidence for an ionic mechanism

The first piece of evidence that must be accounted for is the rate law. For the reaction of benzaldehyde with hydroxide, the reaction is first order with respect to hydroxide ions and second order with respect to benzaldehyde (third order overall).

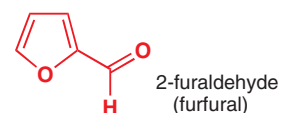
$$\text{rate} = k_4[\text{PhCHO}]^2[\text{HO}^-]$$

For some aldehydes, such as formaldehyde and furfural, the order with respect to the concentration of hydroxide varies between one and two depending on the exact conditions. In high concentrations of base it is fourth order.

$$\text{rate} = k_4[\text{RCHO}]^2[\text{HO}^-]^2$$

➡ The Cannizzaro reaction first appeared in Chapter 26.

➡ For some examples of radical initiators and inhibitors, see Chapter 37. Radical inhibitors are usually stable radicals, such as those on p. 975.



At lower concentrations of base the rate law is a mixture of both third- and fourth-order terms.

$$\text{rate} = k_3[\text{RCHO}]^2[\text{HO}^-] + k_4[\text{RCHO}]^2[\text{HO}^-]^2$$

➡ See p. 261 for an explanation of this important point.

Just because the overall order of reaction is third or fourth order, it does not mean that all the species must simultaneously collide in the rate-determining step. You saw in Chapter 12 that the rate law actually reveals all the species that are involved *up to and including* the rate-determining step.

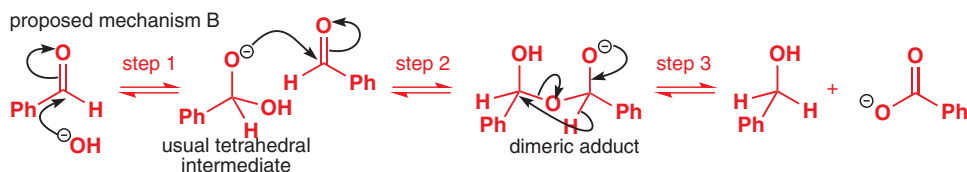
Isotopic labelling

When the reaction is carried out in D_2O instead of in H_2O it is found that there are no C–D bonds in the products. This tells us that the hydrogen must come from the aldehyde and not from the solvent.

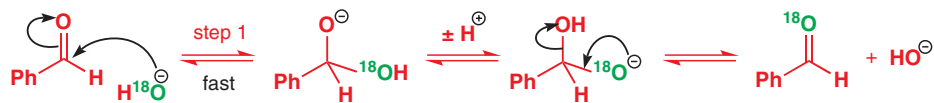


Proposed mechanism B: formation of an intermediate dimeric adduct

A possible mechanism that fits all the experimental evidence so far involves nucleophilic attack of the usual tetrahedral intermediate on another aldehyde to give an intermediate adduct. This adduct could then form the products directly by hydride transfer. You may not like the look of this last step, but the mechanism was proposed and evidence is needed to disprove it.

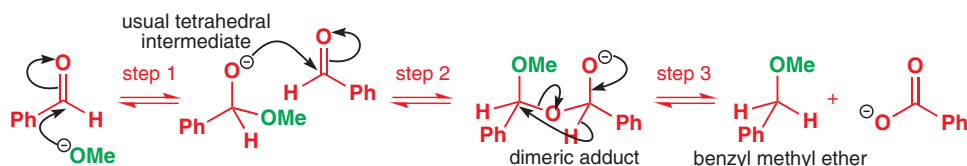


Which step would be rate determining for this mechanism? It could not be step 1 since, if this were the case, the rate law would be first order with respect to the aldehyde rather than the observed second-order relationship. Also, if the reaction is carried out in water labelled with oxygen-18, the oxygen in the benzaldehyde exchanges with the ^{18}O from the solvent much faster than the Cannizzaro reaction takes place. This can only be because of a *rapid* equilibrium in step 1 and so step 1 cannot be rate determining.



So, for mechanism B, either step 2 or step 3 could be rate determining—either case would fit the observed rate law. Step 2 is similar to step 1: in both cases an oxyanion nucleophile attacks the aldehyde. Since the equilibrium in step 1 is very rapid, it is reasonable to suggest that the equilibrium in step 2 should also be rapid and thus that the hydride transfer in step 3 must be rate determining. So mechanism B can fit the rate equation.

How can mechanism B be ruled out? One way is to change the attacking nucleophile. The Cannizzaro reaction works equally well if methoxide is used in a mixture of methanol and water. If mechanism B were correct, the reaction with methoxide would be as follows.



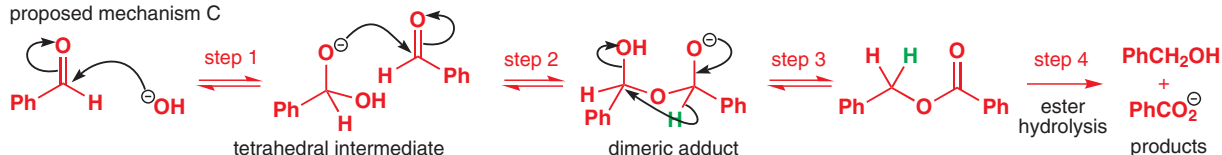
One of the products would be different by this mechanism: benzyl methyl ether would be formed instead of benzyl alcohol. None is observed experimentally. Moreover, under the conditions of the experiment, benzyl methyl ether does not react to form benzyl alcohol, so it cannot be the case that the ether is formed but then reacts to form the products. Mechanism B can therefore be ruled out.

➡ We shall discuss this kind of technique as well as other evidence used to evaluate an intermediate towards the end of this chapter.

Proposed mechanism C: formation of an ester intermediate

This mechanism is like mechanism B but the hydride transfer in the adduct formed in step 2 displaces OH^- to form an ester (benzyl benzoate) that is then hydrolysed to the products. This was at one time held to be the correct mechanism for the Cannizzaro reaction. One piece of evidence for this, and at first glance a very good one, is that by cooling the reaction mixture and avoiding excess alkali, some benzyl benzoate could be isolated during the reaction. An important point is that this does not mean that the ester must be an *intermediate* in the reaction—it might be formed at the end of the reaction, for example. However, it does mean that any mechanism we propose must be able to account for its formation. For now though we want to try to establish whether the ester is an intermediate rather than a by-product in the Cannizzaro reaction.

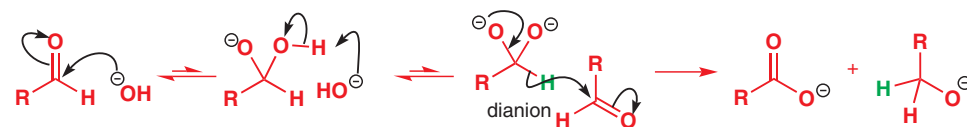
proposed mechanism C



An early objection to mechanism C was that the ester would not be hydrolysed fast enough. When someone actually tried it under the conditions of the experiment, they found that benzyl benzoate is very rapidly hydrolysed (the moral here is 'don't just think about it, try it!'). However, just because the ester *could be* hydrolysed, it still did not show that it actually was an intermediate in the reaction. How this was eventually shown was rather clever. The argument goes like this. We can measure the rate constant for step 4 by seeing how quickly pure benzyl benzoate is hydrolysed to benzyl alcohol and benzoate under the same conditions as those of the Cannizzaro reaction. We also know how quickly these products are formed during the Cannizzaro reaction itself. Since, if this mechanism is correct, the only way the products are formed is from this intermediate, it is possible to work out how much of the intermediate ester must be present at any time to give the observed rate of formation of the products. If we can measure the amount of ester that is actually present and it is significantly less than that which we predict, then this cannot be the correct mechanism. It turned out that there was never enough ester present to account for the formation of the products in the Cannizzaro reaction and mechanism C could be ruled out.

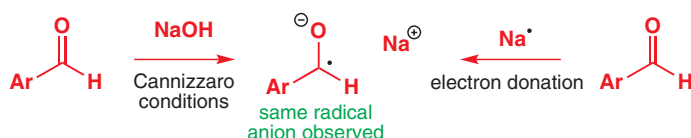
The correct mechanism for the Cannizzaro reaction

The only mechanism that has not been ruled out and that appears to fit all the evidence is the one we have already given (p. 1031). The fact that the rate law for this mechanism is overall third and sometimes fourth order depending on the aldehyde and the conditions can be explained by the involvement of a second hydroxide ion deprotonating the tetrahedral intermediate to give a dianion. When methoxide is used in a methanol/water mix, some methyl ester is formed. This does not stay around for long—under the conditions of the experiment it is quickly hydrolysed to the carboxylate.



Even this mechanism does not quite fit all the evidence

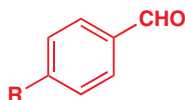
We said earlier that we can never prove a mechanism—only disprove it. Unfortunately, just as the ‘correct’ mechanism seems to be found, there are some observations that make us doubt this mechanism. In Chapter 37 you saw how a technique called **electron spin resonance (ESR)** (or electron paramagnetic resonance, EPR) detects radicals and gives some information about their structure. When the Cannizzaro reaction was carried out with benzaldehyde and a number of substituted benzaldehydes in an ESR spectrometer, a radical was detected. For each aldehyde used, the ESR spectrum proved to be identical to that formed when the aldehyde was reduced using sodium metal. The radical formed was the radical anion of the aldehyde.



Our mechanism does not explain this result, but small amounts of radicals are formed in many reactions in which the products are actually formed by simple ionic processes. Detection of a species in a reaction mixture does not prove that it is an intermediate. Few chemists think that radicals are involved in the Cannizzaro reaction. Most think the mechanism we have given is correct.

Variation in the structure of the aldehyde

Before leaving the Cannizzaro reaction, look at these rates of reactions for aromatic aldehydes with different substituents in the *para* position. These aldehydes may be divided into two classes: those that react faster than unsubstituted benzaldehyde and those that react more slowly. Those that go slower all have something in common—they all have substituents on the ring that donate electrons.



Rate of Cannizzaro reaction with aromatic aldehydes

R =	Rate relative to benzaldehyde at 25°C	Rate relative to benzaldehyde at 100°C
H	1	1
Me	0.2	0.2
MeO	0.05	0.1
Me ₂ N	very slow	0.0004
NO ₂	210	2200

We have already seen how substituents on a benzene ring affect the rate of electrophilic substitution (Chapter 21). Electron-donating groups such as MeO and Me₂N dramatically speed up the rate at which an aromatic ring is attacked by an electrophile, whereas electron-withdrawing groups, particularly nitro groups, slow the reaction down. The Cannizzaro reaction is not taking place on the benzene ring itself, but substituents on the ring still make their presence known. The fact that the Cannizzaro reaction goes much *slower* with electron-donating groups and faster with electron-withdrawing groups tells us that, for this reaction, rather than a positive charge developing, as in the case of electrophilic substitution on an aromatic ring, there must be negative charge accumulating somewhere near the ring. The accumulation of more negative charge is disfavoured by the presence of a group that is already offloading electron density into the ring. In agreement with this, our mechanism has mono- and dianion intermediates, which are stabilized by electron-withdrawing groups and destabilized by electron-donating groups.

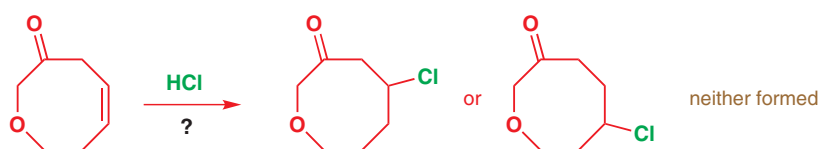
➡ Later in the chapter you will see a more quantitative treatment of this variation of structure.

The rest of the chapter is devoted to discussions of methods similar to those we have briefly surveyed for the Cannizzaro reaction, with examples of the use of each method. You can assume that the mechanisms we have discussed in this book have been verified (not, of course, proved) by these sorts of methods.

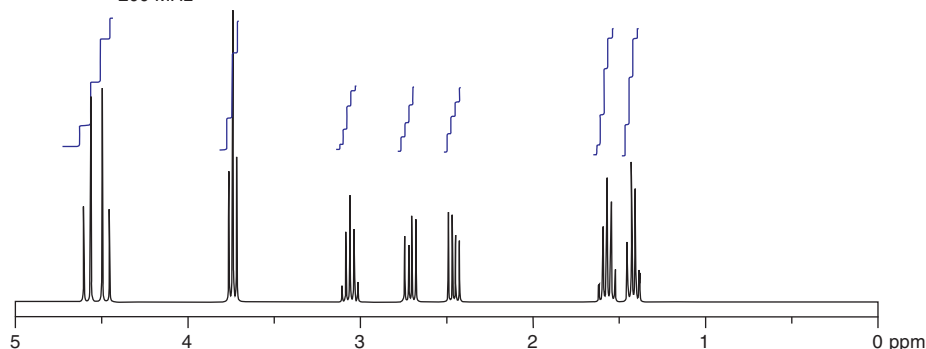
Be sure of the structure of the product

This seems a rather obvious point. However, there is a lot to be learned from the detailed structure of the product: its connectivity (which atom goes where) as well as its stereochemistry. You will see that it may be necessary to alter the structure of the starting material in subtle ways to make sure that we know exactly what happens to all its atoms by the time it reaches the product.

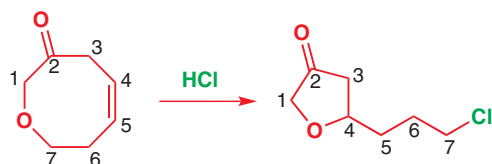
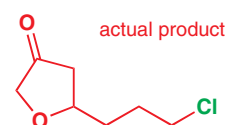
Suppose you are studying the addition of HCl to this alkene. You find that you get a good yield of a single adduct and you might be a bit surprised that you do not get a mixture of the two obvious adducts. You may wonder if there is some participation of the ether oxygen or whether perhaps the ketone enolizes during the reaction and controls the outcome.



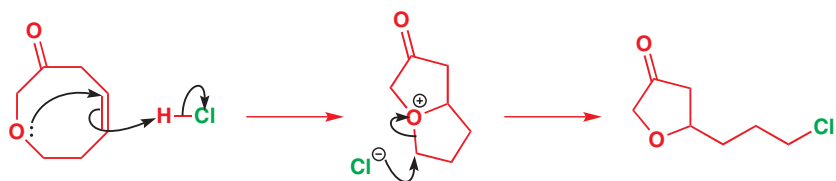
1H NMR spectrum of product
200 MHz



If you are cautious you might check on the structure of the product before you start a mechanistic investigation. The NMR spectrum (above) tells you at once that the product is neither of these suggestions. It contains a $(CH_2)_3Cl$ unit and can no longer have an eight-membered ring. A ring contraction has given a five-membered ring and a mechanistic investigation is hardly needed. Simply knowing what the product is allows us to propose a mechanism. A rearrangement has occurred and we could use the method suggested in Chapter 36: number the atoms in the starting material and find them in the product. This is quite easy as only one numbering system makes any sense.



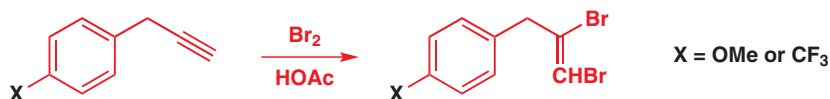
This numbering suggests that the carbon skeleton is unaffected by the reaction, that protonation has occurred at C5, that the ether oxygen has acted as an internal nucleophile across the ring at C4, and that the chloride ion has attacked C7. The mechanism is straightforward.



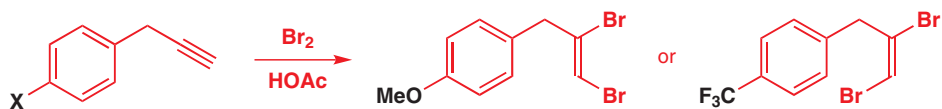
It may be disappointing to find that every step in this mechanism is well known and that the reaction is exactly what we ought to have expected with an eight-membered ring as these rings are famous for their transannular (across-ring) reactions to form 5/5 fused systems. However, it is good that a prolonged investigation is not necessary.

● Find out for sure what the structure of the product is before you start a mechanistic investigation.

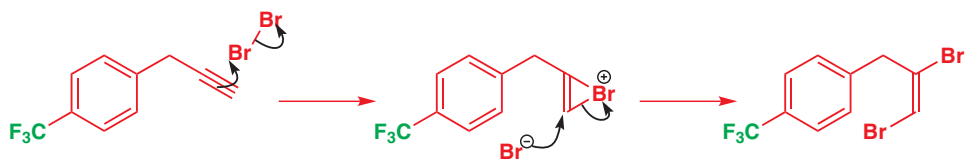
A more subtle distinction occurred in a study of the bromination of alkynes. Bromination of benzyl alkynes in acetic acid gave the products of addition of one molecule of bromine—the 1,2-dibromoalkenes. The reaction was successful with a variety of *para* substituents and there seems at first to be no special interest in the structure of the products.



Closer investigation revealed an extraordinary difference between them, not at all obvious from their NMR spectra: the compound from X=OMe was the *Z*-dibromoalkene from *cis* addition of bromine while the product from X=CF₃ was the *E* alkene from *trans* addition. What mechanism could explain this difference?

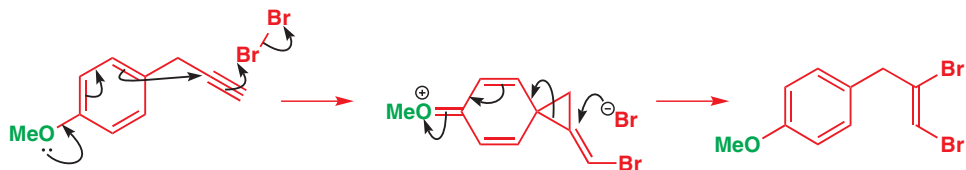


The *anti* addition is more easily explained: it is the result of formation of a bromonium ion, similar, in fact, to the normal mechanism for the bromination of alkenes. Bromine adds from one side of the alkene and the bromide ion must necessarily form the *E*-dibromo product regardless of which atom it attacks.



So why does the *p*-methoxy-substituted compound behave differently? It cannot react by the same mechanism and a reasonable explanation is that the much more electron-donating ring participates in the reaction to give a carbocyclic three-membered ring intermediate that is attacked in an *anti* fashion to give the *Z* alkene. Both intermediates are three-membered ring cations and both are attacked with inversion but the *p*-MeO compound undergoes double inversion by participation of the ring.

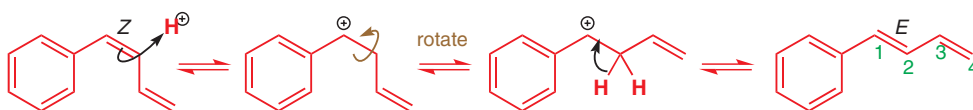
➡ A similar aryl participation in saturated compounds, giving a 'phenonium ion' intermediate, appears in Chapter 36, p. 936.



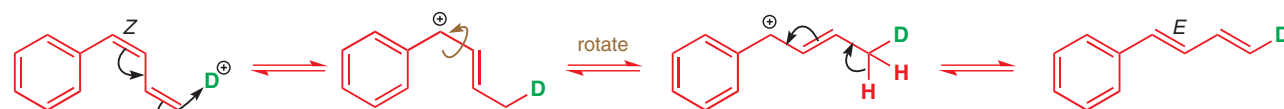
Labelling experiments reveal the fate of individual atoms

It often happens that the atoms in starting material and product cannot be correlated without at least one of them being labelled. The fact that many elements exist as different isotopes provides us with a perfect way of doing this: a neutron more or less in the nucleus affects the physics (and hence the spectroscopic features) of an atom, but not its chemistry.

The isomerization of *Z*-1-phenylbutadiene to the *E* diene in acid looks like a simple reaction. Protonation of the *Z* alkene would give a stabilized secondary benzylic cation that should last long enough to rotate. Loss of the proton would then give the more stable *E* diene.

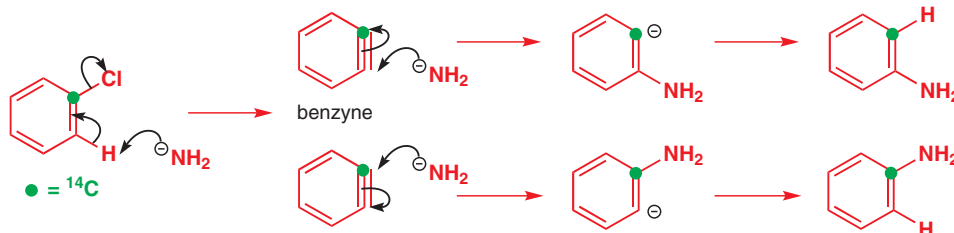


However, reaction with D^+ in D_2O reveals that this mechanism is incorrect. The product contains substantial amounts of deuterium at C4, not at C2 as predicted by the proposed mechanism. Protonation must occur at the end of the conjugated system to produce the more stable conjugated cation, which rotates about the same bond and loses H or D from C4 to give the product. More H than D will be lost, partly because there are two Hs and only one D, but also because of the kinetic isotope effect, of which more later.

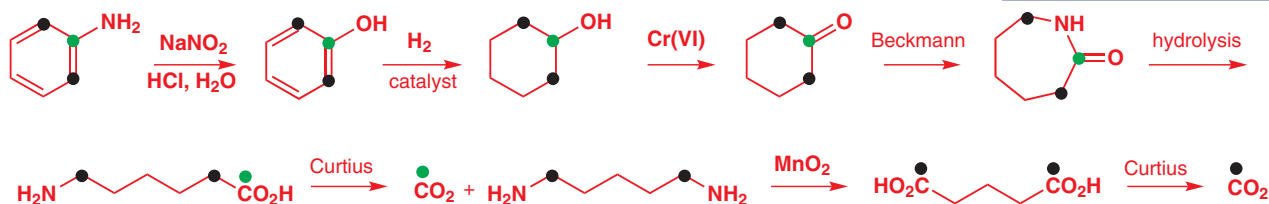


The easiest labels to use for this job are D for H, ^{13}C , and ^{18}O . None of these is radioactive; all can be found by mass spectrometry, while D and ^{13}C can be found by NMR. Older work on mechanisms used radioactive tracers such as T (tritium, 3H) for H and ^{14}C .

The first evidence for benzyne as the intermediate in the reaction of chlorobenzene with NH_2^- came from radioactive labelling. If benzyne is an intermediate, the product should have 50% label at C1 and 50% at the two identical *ortho* carbons, as the scheme below shows.



The labelled aniline was degraded by the reactions shown here, which you must agree was a lot of work for the chemists concerned. Each potentially labelled carbon atom had to be isolated from any other labelled atom and the radioactivity measured. We shall follow the fate of the two labelled atoms with black and green spots. Since the two *ortho* positions are identical, we must put a black spot on both of them.



Most of these reactions are well known—the Beckmann rearrangement is described in Chapter 36 and the Curtius reaction in Chapter 38—but the oxidation of the diamine to the

■ In fact a feature known as the *kinetic isotope effect* means that isotopes of an element can have subtly different chemistry, as we will explain on p. 1050.

■ Radioactive isotopes are, of course, more dangerous to use but they can at least always be found. The real disadvantage is that, to discover exactly where they are in the product, the molecule must be degraded in a known fashion. Radioactive isotopes are rarely used now except in determining biological mechanisms, as you will see in Chapter 42. Tritium and ^{14}C are β emitters—they give off electrons—having half-lives of 12 and over 5000 years, respectively. Tritium is made on a large scale by neutron irradiation of 6Li in a nuclear reactor.

Interactive mechanism showing benzyne intermediate

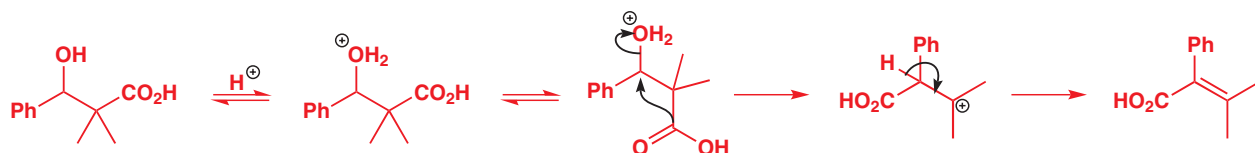
➡ Benzyne is discussed in Chapter 22 as an intermediate in nucleophilic aromatic substitution.

Other symmetrical intermediates originally identified by radioactive labelling include the cyclopropanone in the Favorskii rearrangement in Chapter 36, p. 950, and a spirocyclic intermediate in electrophilic substitution on an indole in Chapter 29, p. 746.

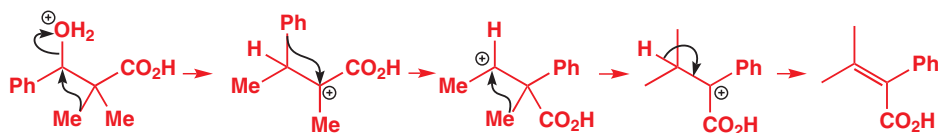
dicarboxylic acid is not a standard procedure and is not recommended. All the label came out in the CO_2 and almost exactly half of it was from the black and half from the green labelled carbons. This was the original evidence that convinced organic chemists in 1953 that benzyne was involved in the reaction. The evidence presented in Chapter 22 is much more modern.

The value of double labelling experiments

An altogether more modern approach to a labelling study was used in the surprising rearrangement of a hydroxy-acid in acidic solution. The structure of the product suggests a CO_2H migration as the most likely mechanism. This mechanism resembles closely the cationic rearrangements of Chapter 36.

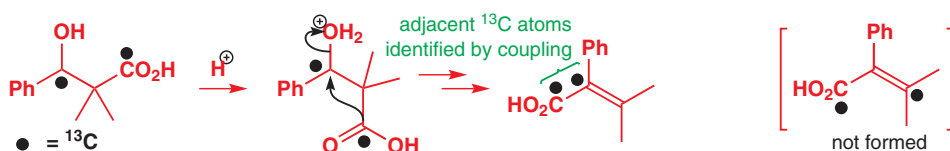


But there are other possibilities: received wisdom (Chapter 36) suggests that the best migrating group in cationic rearrangements is the one best able to bear a positive charge, by which logic the more familiar Ph and Me migrations ought to be preferred. A more elaborate mechanism can be written: it involves two methyl migrations and one phenyl migration and it also needs consideration.



We don't normally see ^{13}C – ^{13}C coupling because in most molecules only 1.1% of the C atoms are ^{13}C , so there's little chance of two of them finding themselves adjacent to one another. But when both C atoms are always ^{13}C , the coupling becomes evident.

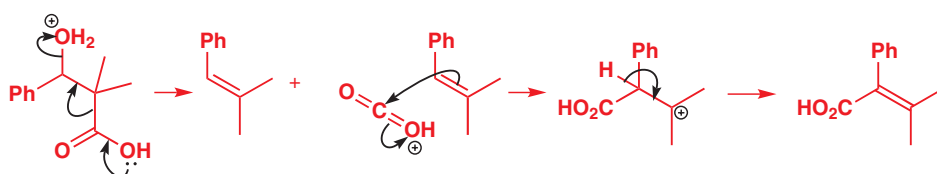
These mechanisms can be tested by finding out whether the CO_2H group remains attached to its original position or becomes attached to the other carbon in the skeleton of the molecule. This can be done by double labelling. If a compound is prepared with two ^{13}C labels, one on the CO_2H group itself and one on the benzylic carbon, the NMR spectrum of the product will show what has happened. In fact, the two ^{13}C labels end up next to each other with a coupling constant $^1J_{\text{CC}} = 71 \text{ Hz}$. It is the CO_2H group that has migrated.



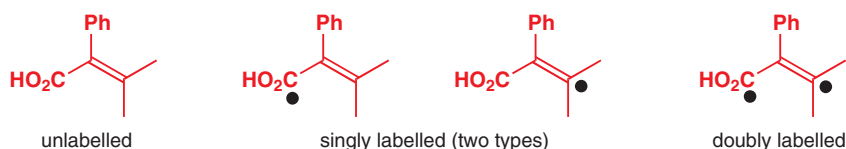
So why does the CO_2H group migrate? It does so not because it is a good migrating group, but because it cannot bear to be left behind. The rearranged cation from CO_2H migration is a stable tertiary alkyl cation. The cation from Me migration is a very unstable cation, with the positive charge next to the CO_2H group. Such cations are unknown as the carbonyl group is very electron withdrawing.

'Crossover' experiments

There is still one tiny doubt. Supposing the reaction is not intramolecular at all, but *intermolecular*. The CO_2H group might be lost from one molecule as protonated CO_2 and be picked up by another molecule of alkene. No migration would be involved at all.



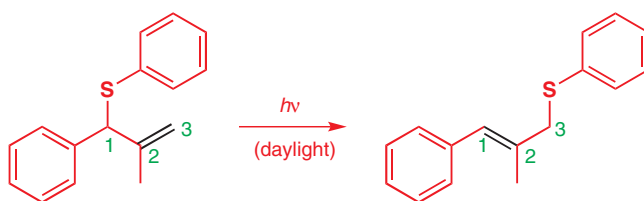
This mechanism can be checked by using a 50:50 mixture of doubly labelled and unlabelled starting material. The molecule of alkene that captures the roving protonated labelled CO_2 might happen to be labelled too but equally well it might be unlabelled. If this last mechanism is correct, we should get a mixture of unlabelled, singly labelled, and doubly labelled product in the ratio 1:2:1 as there are two types of singly labelled product. The two singly labelled compounds are called the crossover products and the experiment is called a crossover experiment as it discovers whether any parts of one molecule cross over to another.



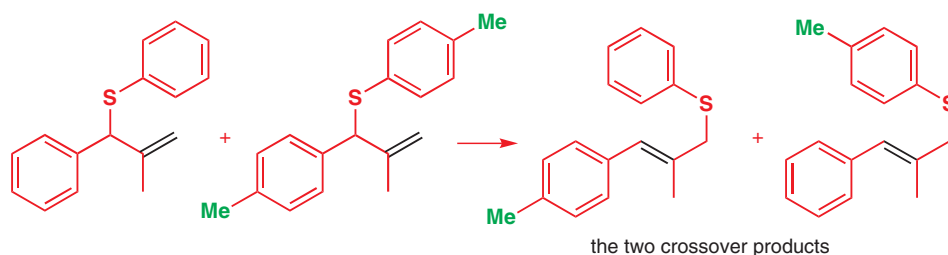
➡ There is an example of a crossover experiment proving that an $\text{S}_{\text{N}}2$ reaction is intermolecular in Chapter 31, p. 811 and one proving the mechanism of a rearrangement on pp. 959–960.

In fact, no singly labelled compounds were found: NMR analysis showed that the product consisted entirely of unlabelled or doubly labelled molecules. The CO_2H group remains attached to the same molecule (though not to the same atom) and the first mechanism is correct.

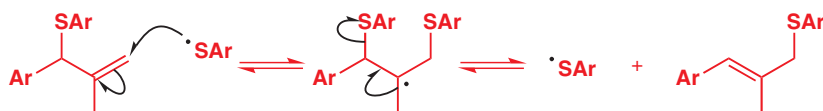
Crossover experiments demand some sort of double labelling, which does not have to be isotopic. An example where crossover products are observed is the light-initiated isomerization of allylic sulfides.



This is formally a [1,3] sigmatropic shift of sulfur (Chapter 35) but that is an unlikely mechanism (and you should be able to suggest why). A crossover experiment was carried out in which the two molecules had either two phenyl groups or two *para*-tolyl groups. The mixture was allowed to rearrange in daylight and the products were examined by mass spectroscopy. There was a roughly 1:2:1 mixture of products having two phenyl groups, one phenyl and one *para*-tolyl group, and two *para*-tolyl groups. The diagram shows the starting materials and the two crossover products only.



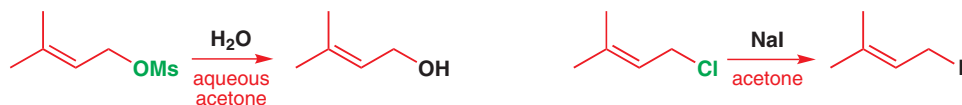
Clearly, the ArS group had become separated from the rest of the molecule and the most likely explanation was a radical chain reaction (Chapter 37) with the light producing a small amount of $\text{ArS}^\bullet$ to initiate the chain. The *para*-methyl group acts as a label. The whole system is in equilibrium and the more highly substituted alkene is the product.



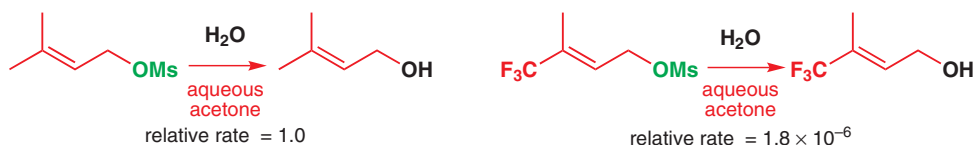
Systematic structural variation

In this last example, the hope is that the *para*-methyl group will have too weak an electronic or steric effect and in any case will be too far away to affect the outcome. It is intended to make nearly as slight a change in the structure as an isotopic label. Many structural investigations have exactly the opposite hope. Some systematic change is made in the structure of the molecule in the expectation of a predictable change in rate. A faster or slower reaction will lead to some definite conclusion about the charge distribution in the transition state.

Allylic compounds can react efficiently with nucleophiles by either the S_N1 or S_N2 mechanisms (Chapter 15). Here are two examples.



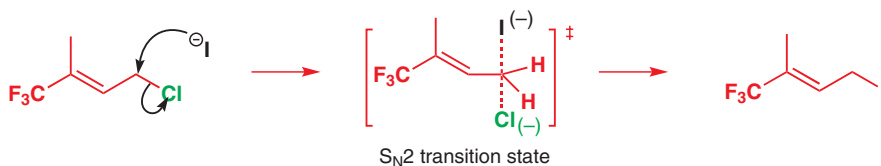
The carbon skeleton is the same in both reactions but the leaving groups and the nucleophiles are different. These reactions might both go by S_N1 or S_N2 or one might go by S_N1 and the other by S_N2 . One way to find out is to make a large change in the electronic nature of the carbon skeleton and see what happens to the rate of each reaction. In these experiments one of the methyl groups was changed for a CF_3 group—exchanging a weakly electron-donating group for a strongly electron-withdrawing group. If a cation is an intermediate, as in the S_N1 reaction, the fluorinated compound will react much more slowly. Here is the result in the first case.



The fluorinated compound reacts half a million times more slowly so this looks very much like an S_N1 mechanism. The slow step in an S_N1 mechanism is the formation of a carbocation so any group that destabilizes the positive charge would have (and evidently does have) a large effect on the rate. Rate ratios of several powers of ten often are worth noticing and a rate ratio of nearly 10^6 is considerable. In the second case the rate difference is much less.



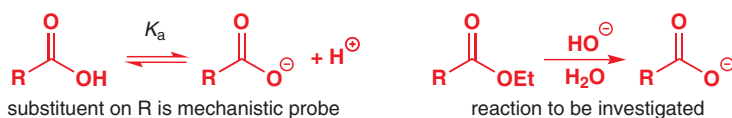
A rate ratio of 11 is not worth noticing. The point is not that the fluorinated compound reacts faster but that the two compounds react at about the same rate. This strongly suggests that no charge is generated in the transition state and an S_N1 mechanism is not happening. The S_N2 mechanism makes good sense with its concerted bond formation and bond breaking requiring no charge on the carbon skeleton.



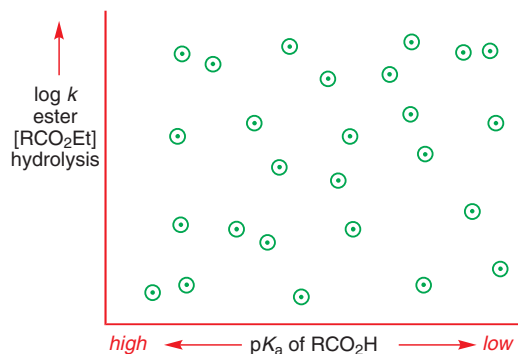
The CF_3 group works well here as a mechanistic probe because it is held well out of the way of the reaction site by a rigid π system but is connected electronically by that same allylic system. Steric effects should be minimized and electronic effects clearly seen. This approach is clearly limited by the small number of groups having properties like those of the CF_3 group and the small number of reactions having such favourable carbon skeletons. We will now present the most important serious correlation between structure and reactivity.

The Hammett relationship

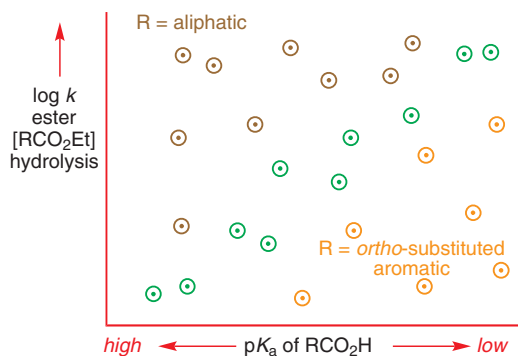
What we would ideally like to do is find a way to quantify the effects that electron-donating or -withdrawing groups have on the transition state or intermediate during the course of a reaction. This will then give us an idea of what the transition state is really like. The first question is: can we define exactly how efficient a given group is at donating or withdrawing electrons? Hammett took the arbitrary decision to use the pK_a of an acid as a guide. For example, the rate of hydrolysis of esters might well correlate with the pK_a of the corresponding acid.



When Hammett plotted the rates of ethyl ester hydrolyses (as $\log k$ since pK_a has a log scale) against the pK_a s of the corresponding acids, the initial results were not very encouraging as there was a random scatter of points over the whole graph.



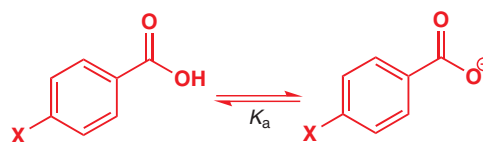
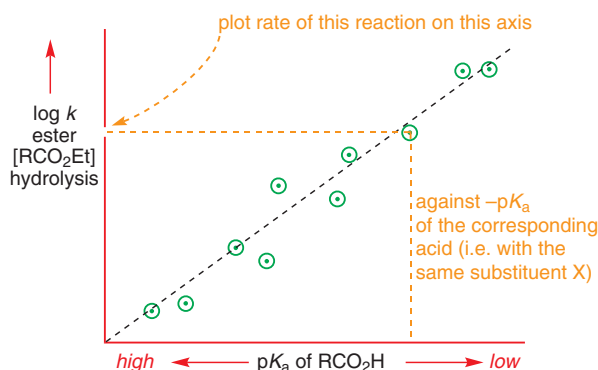
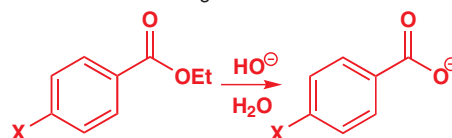
Hammett had used some aliphatic acids (substituted acetic acids) and some aromatic acids (substituted benzoic acids) and he noticed that many of the points towards the top of the graph belonged to the substituted acetic acids. Removing them (brown points) made the graph a lot better. He then noticed that the remaining aromatic compounds were in two classes: the *ortho*-substituted esters reacted more slowly than their *meta*- and *para*-isomers and came towards the bottom of the graph (orange points). Removing them made the graph quite good (remaining green points).



It was not a perfect correlation but Hammett had removed the examples where steric hindrance was important. Aliphatic compounds can adopt a variety of conformations (Chapter 16) and the substituent in some of them will interfere with the reaction. Similarly, in *ortho*-substituted aromatic compounds the nearby substituent might exert steric hindrance on the reaction. Only with *meta*- and *para*-substituted compounds was the substituent held out of the way, on a rigid framework, and in electronic communication with the reaction site through the flat but conjugated benzene ring.

Notice that the straight line is not perfect. This graph is an invention of the human mind. It is a correlation between things that are not directly related. If you determine a rate constant by plotting the right function of concentration against time and get an imperfect straight line, that is your fault because you haven't done your measurements carefully enough. If you make a Hammett plot and the points are not on a straight line (and they won't be) then that is *not* your fault. The points really don't fit on a perfectly straight line. As you will see soon, this doesn't actually matter.

reaction to be investigated



Louis P. Hammett (1894–1987) invented 'physical organic chemistry' and at Columbia University in 1935 derived the Hammett σ/ρ relationship. The impact was enormous and in the 1960s chemists were still working out more such correlations.

The Hammett substituent constant σ

■ You cannot push arrows from the negative charge of the carboxylate anion into the ring. Try it.

A quick glance at the pK_a s of some substituted benzoic acids in the table below will show how well they correlate electron donation with pK_a . The substituents at the top of the table are electron donating and the anions of the benzoic acids are correspondingly less stable so these are the weakest acids. At the bottom of the table we have the electron-withdrawing groups, which stabilize the anion and make the acid stronger. The whole range is not that great, only one pH unit or so, because the carboxylate anion is not conjugated with the ring.

Hammett decided not to use the pK_a s themselves for his correlation but defined a new parameter, which he called σ . This σ shows how electron donating or withdrawing a group is relative to H as a difference between the pK_a s of a benzoic acid derivative with the substituent and benzoic acid itself. If the acid required to determine σ for a new substituent was not available, σ could be determined by correlation with other reactions. Here are the equations and the table of σ values for the most important substituents. A different value of σ for any given substituent was needed for the *meta* and the *para* positions and these are called σ_m and σ_p , respectively.

$$\sigma_x = \log \left(\frac{K_a(\text{XC}_6\text{H}_4\text{COOH})}{K_a(\text{C}_6\text{H}_5\text{COOH})} \right) = pK_a(\text{C}_6\text{H}_5\text{COOH}) - pK_a(\text{XC}_6\text{H}_4\text{COOH})$$

Substituent X	pK_a of $p\text{-XC}_6\text{H}_4\text{COOH}$	pK_a of $m\text{-XC}_6\text{H}_4\text{COOH}$	σ_p	σ_m	Comments
NH ₂	4.82	4.20	-0.62	0.00	groups that donate electrons have negative σ
OCH ₃	4.49	4.09	-0.29	0.11	
CH ₃	4.37	4.26	-0.17	-0.06	there are no values for <i>ortho</i> substituents
H	4.20	4.20	0.00	0.00	
F	4.15	3.86	0.05	0.34	
I	3.97	3.85	0.23	0.35	
Cl	3.98	3.83	0.22	0.37	$\sigma_p < \sigma_m$ for inductive withdrawal
Br	3.97	3.80	0.23	0.40	
CO ₂ CH ₃	3.75	3.87	0.45	0.33	$\sigma_p > \sigma_m$ for conjugating substituents
COCH ₃	3.71	3.83	0.49	0.37	
CN	3.53	3.58	0.67	0.62	groups that withdraw electrons have positive σ
NO ₂	3.43	3.47	0.77	0.73	

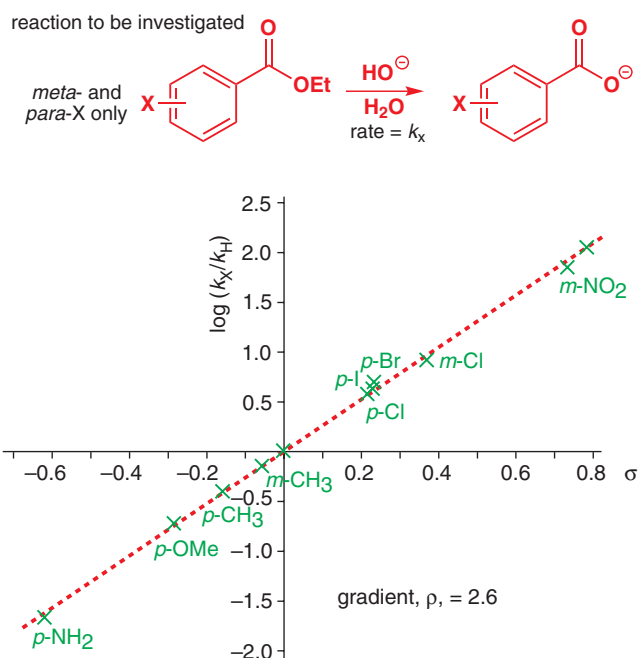
There is no point learning the precise figures in this table, but it will help if you form a general idea of what a σ value means. If $\sigma = 0$ the substituent has no effect: it is electronically the same as H. If σ is positive, the substituent is electron withdrawing. This is unfortunate perhaps, but just remember that the comparison is with acid strength, and acids with electron-withdrawing substituents are stronger. Positive σ means a stronger acid so the substituent is electron withdrawing. The more positive the charge induced on the ring by a substituent, the larger its σ value. Negative σ means weaker acid and electron donation. Inductive effects from polarization of σ bonds are greater for σ_m than for σ_p because the substituent is nearer.

Conjugation is generally more effective in the *para* position (see Chapter 21) so $\sigma_p > \sigma_m$ for conjugating substituents. Indeed, the NH_2 group has a large negative σ_p and a zero σ_m . The NH_2 group donates electrons strongly to the carbonyl group of benzoic acid from the *para* position but does not conjugate in the *meta* position where its donation happens just to balance the effect of electronegative nitrogen.

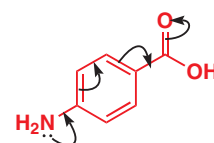
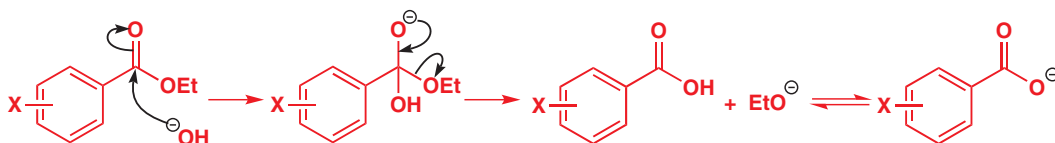
The OMe group has a negative σ_p but a positive σ_m because a weaker electron donation from the lone pairs is more important in the *para* position but the effect of very electronegative oxygen on the σ framework of the ring in the *meta* position is more important than lone pair donation that doesn't reach the carbonyl group. You do not need to learn any σ values but you should be able to work out the sign of σ for well-known substituents and estimate a rough value.

The Hammett reaction constant ρ

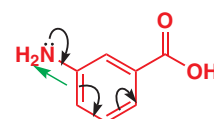
Now we can return to our reaction: the alkaline hydrolysis of various *meta*- and *para*-substituted ethyl benzoates. We sketched the graph earlier (p. 1041) but now we can add some more quantitative detail. The rate constants for this second-order reaction have been measured and shown here is a graph of $\log(k_X/k_H)$ versus σ , where k_X is the rate constant for the reaction with the substituted benzoate and k_H is that for the unsubstituted reaction ($X=\text{H}$).



We can see straight away that there is a good correlation between how fast the reaction goes and the value of σ ; in other words, the points lie more or less on a straight line. The gradient of this best-fit line, given the symbol ρ (rho), tells us how sensitive the reaction is to substituent effects in comparison with the ionization of benzoic acids. The gradient is $\rho = +2.6$. This tells us that the reaction responds to substituent effects in the same way (because it is +) as the ionization of benzoic acids but by much more ($10^{1.6}$ times more) because it is 2.6 instead of 1.0. We already know what the mechanism of this reaction is:



strong conjugation
into carbonyl group:
large negative σ_p



conjugation into ring
not carbonyl group
balances weak effect
of electronegative N:
zero σ_m

Getting to grips with logs

A difference between two values of x log units means the values actually differ by a factor of 10^x . From the graph for the hydrolysis of ethyl benzoates you can see that the *p*- NO_2 benzoate hydrolyses some 10^2 times faster than the unsubstituted benzoate, while the *p*- NH_2 benzoate hydrolyses some 10^2 times slower.

Hammett chose σ (Greek ς) for *substituent* and ρ (Greek ρ) for *reaction*.

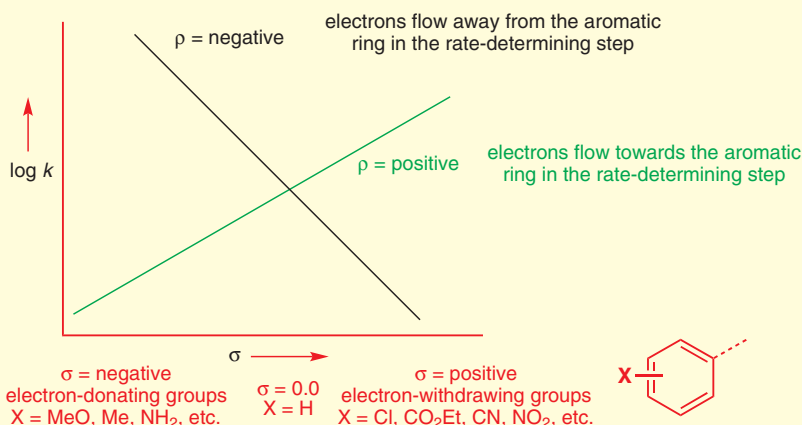
The first step is quite like the ionization of benzoic acid. A negative charge is appearing on the carbonyl oxygen atom and that negative charge will be stabilized by electron-withdrawing X groups. If the first step is rate determining, a positive ρ makes sense.

We need now to look at some other reactions to get a grasp of the meaning of the value of the Hammett ρ .

● **The Hammett reaction constant ρ measures the sensitivity of the reaction to electronic effects.**

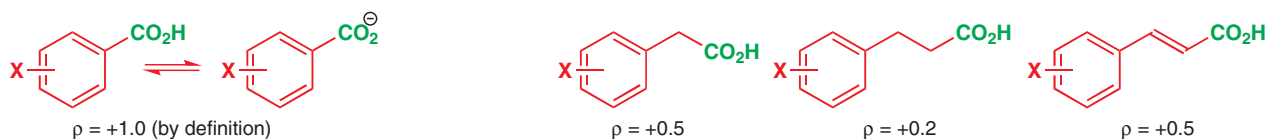
- A **positive ρ value** means **more electrons** in the transition state than in the starting material.
- A **negative ρ value** means **fewer electrons** in the transition state than in the starting material.

typical Hammett plots



Equilibria with positive Hammett ρ values

To take a simple example, let's just see what happens to ρ if we simply move the carboxylic acid away from the ring. The ρ value for ionization gets less. This is just what you would expect—the further it is from the aromatic ring, the less the acid cares about how electron rich or poor the ring is. With two saturated carbons between the benzene ring and the carboxylic acid, there is almost no effect on pK_a . But restore electronic communications with a double bond, and ρ goes back up again.



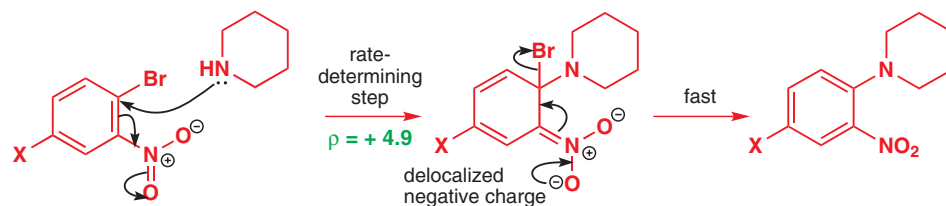
If the negative charge on the anion can actually be delocalized round the ring, as it can in substituted phenols, we should expect the size of ρ to increase. Both the phenol and the anion are delocalized but delocalization is more important for the anion. The effect is even more significant for the ionization of anilinium salts as the acid ArNH_3^+ does not have a delocalized lone pair but the conjugate base (ArNH_2) does.



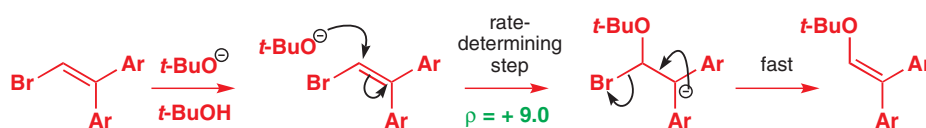
Reactions with positive Hammett ρ values

The size and sign of the value of ρ tell us about what is happening *in the rate-determining step* of a reaction. Any reaction that involves nucleophilic attack on a carbonyl group as the rate-

determining step is going to have a ρ value of about 2–3, the same as for the hydrolysis of esters, as we have already seen. Large positive ρ values usually indicate extra electrons in the transition state delocalized into the ring itself. A classic example is nucleophilic aromatic substitution by the addition–elimination mechanism (Chapter 22). The ρ value is +4.9, but even this large value does not mean a complete anion on the benzene ring as the nitro group, present in all cases, takes most of the negative charge. The substituent X merely helps.

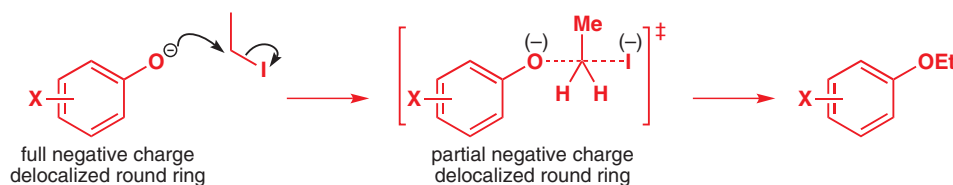


We get the full value when there are no nitro groups to take the brunt of the negative charge. This vinylic substitution has a ρ value of +9.0. It cannot be an S_N2 reaction or it would have a small ρ value and it cannot be an S_N1 reaction or it would have a negative ρ value (fewer electrons in the transition state). It must be an addition–elimination mechanism through a benzylic anion delocalized round both benzene rings.

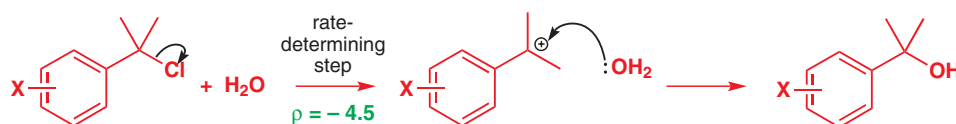


Reactions with negative Hammett ρ values

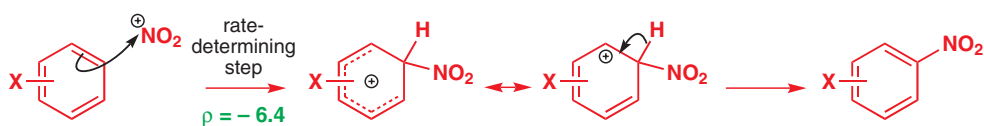
Negative ρ values mean electrons flowing away from the ring. A representative example is the S_N2 displacement of iodide from EtI by phenoxide anions. This has a ρ value of exactly –1.0. Although the transition state has a negative charge, that charge is decreasing on the aromatic ring as the starting material approaches the transition state.



An S_N1 reaction on the carbon atom next to the ring has a large negative ρ value. In this example, a tertiary benzylic cation is the intermediate and the rate-determining step is, of course, the formation of the cation. The cation is next to the ring but delocalized round it and the ρ value is –4.5, about the same value, though negative, as that for the nucleophilic substitution on nitrobenzenes by the addition–elimination mechanism that we saw in the last section.

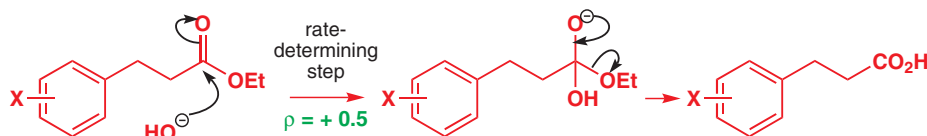


The largest negative ρ values come from electrophilic aromatic substitution (Chapter 21), where the electrons of the ring are used in the reaction, leaving a positive charge on the ring itself in the intermediate. Some of this charge is already there in the transition state. This simple nitration has $\rho = -6.4$ and ρ values for electrophilic aromatic substitution are usually in the range –5 to –9. Negative ρ values mean electrons flowing out of the ring.

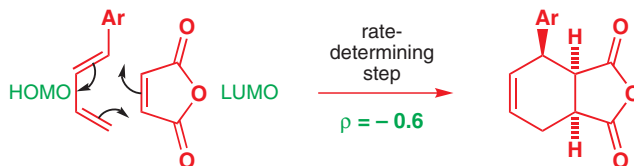


Reactions with small Hammett ρ values

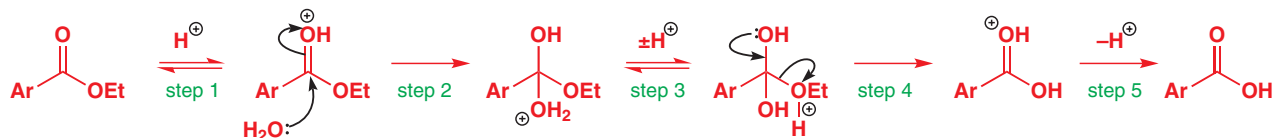
Small Hammett ρ values arise in three ways. The aromatic ring being used as a probe for the mechanism may simply be too far away for the result to be significant. This trivial case of the alkaline hydrolysis of the 3-aryl propionate ester has a ρ value of +0.5 and it is surprising that it is even that large.



The second case is the informative one where the reaction is not dependent on electrons flowing into or out of the ring. Pericyclic reactions are important examples and the Diels–Alder reaction of arylbutadienes with maleic anhydride shows a small negative ρ value of -0.6 . The small value is consistent with a mechanism not involving charge accumulation or dispersal, but the sign is interesting. We explained this type of Diels–Alder reaction in Chapter 34 by using the HOMO of the diene and the LUMO of the dienophile. The negative sign of ρ , small though it is, supports this view because the reaction is somewhat faster with electron-donating groups on Ar, which raise the energy of the HOMO of the diene.



The third case is in many ways the most interesting. We have seen that the alkaline hydrolysis of ethyl esters of benzoic acids (ArCO_2Et) has a ρ value of +2.6 and that this is a reasonable value for a reaction involving nucleophilic attack on a carbonyl group conjugated with the aromatic ring. The hydrolysis of the same esters in acid solution, which also involves nucleophilic attack on the same carbonyl group, has a ρ value of +0.1. In other words, substituted benzoic esters hydrolyse at more or less the same rate in acid solution, irrespective of their substituents. We need to look at the full mechanism to explain this remarkable result.

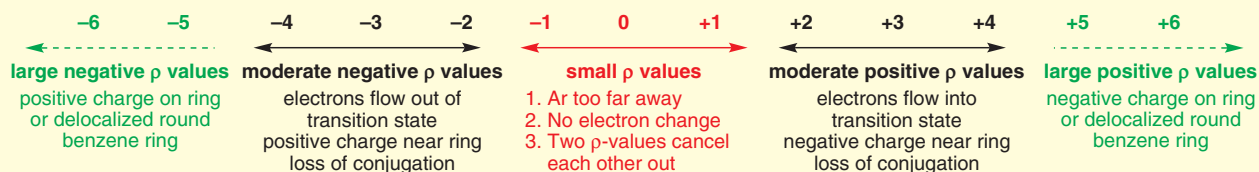


➡ We made this point in Chapter 12, p. 258.

Steps 1, 3, and 5 cannot be slow as they are just proton transfers between oxygen atoms, and proton transfer between electronegative atoms is always fast. That leaves only steps 2 and 4 as possible rate-determining steps. The bimolecular addition of the weak nucleophile water to the low concentration of protonated ester (step 2) is the most attractive candidate, as step 4—the unimolecular loss of ethanol and re-formation of the carbonyl group—should be fast. What ρ value would be expected for the reaction if step 2 were the rate-determining step? It would be made up of two parts. There would be an equilibrium ρ value for the protonation step and a reaction ρ value for the addition of water. Step 1 involves electrons flowing out of the molecule and step 2 involves electrons flowing in so the ρ values for these two steps would

have opposite charges. We know that the ρ value for step 2 would be about +2.5 (it's just like the step in the ester hydrolysis) and a value of about -2.5 for the equilibrium protonation is reasonable. This is indeed the explanation: step 2 is the rate-determining step and the ρ values for steps 1 and 2 almost cancel each other out. All steps before the rate-determining step are present in the rate equation and also affect the Hammett ρ value.

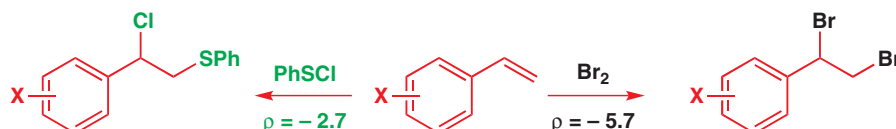
Summary: interpreting Hammett ρ values



You should not, of course, learn the numbers in this scheme, but you need an idea of roughly what each group of values means. You should see now why it is unimportant whether the Hammett correlation gives a good straight line or not. We just want to know whether ρ is + or - and whether it is, say, 3 or 6. It is meaningless to debate the significance of a ρ value of 3.4 as distinct from one of 3.8.

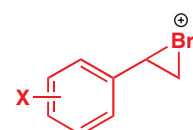
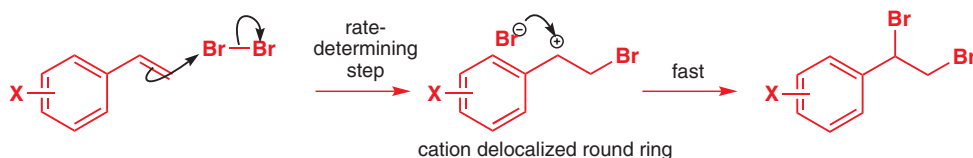
Using the Hammett ρ values to uncover mechanisms

Electrophilic attack on alkenes by bromine often goes through three-membered ring cyclic bromonium ions and we can sometimes tell that this is so by studying the stereochemistry. Here are two reactions of styrenes that look very similar—a reaction with bromine and another with PhSCl. With no further information, we might be tempted to assume that they both go by the same mechanism. However, the Hammett ρ values for the two reactions are rather different.

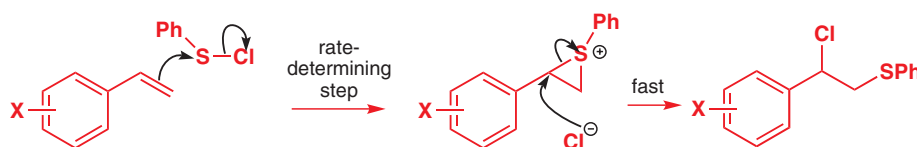


➡ Chapter 19 gives a full description of these mechanisms. There is more about these sulfonyl chlorides in Chapter 27, p. 658.

The ρ value for bromination is definitely in the 'large' range and can only mean that a positive charge is formed that is delocalized round the benzene ring. Bromine evidently does not form a bromonium ion with these alkenes but prefers to form a secondary benzylic cation instead, which can be stabilized more effectively by delocalization.

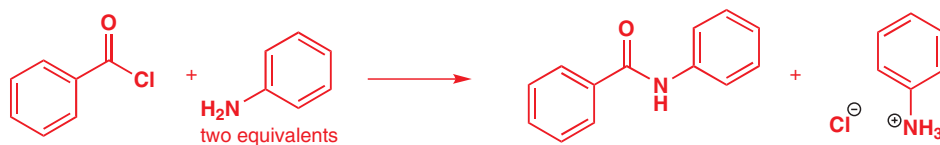


The sulfonylation, on the other hand, has a moderate negative ρ value. No cation is formed that is delocalized round the ring, but electrons flow out of the ring and we suspect some loss of conjugation. All this fits well with the formation of a three-membered ring intermediate. From experiments like this we learn that PhSCl is much more likely than bromine to react stereospecifically with alkenes through cyclic cation intermediates.

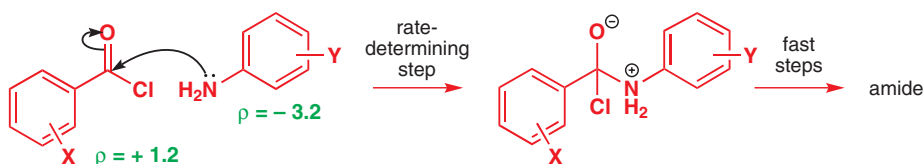


A complete picture of the transition state from Hammett plots

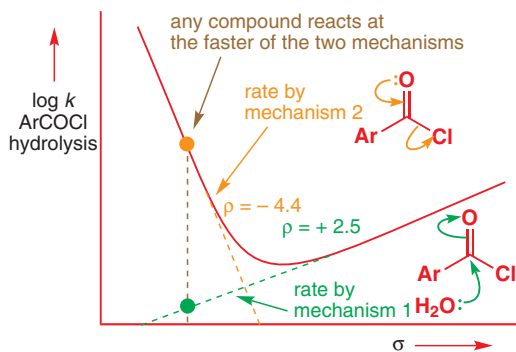
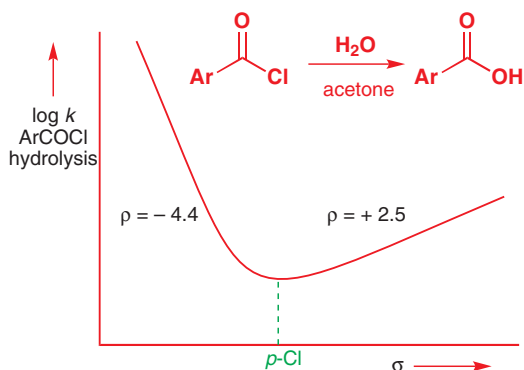
More information can be gained on the mechanism of the reaction if two separate experiments can be carried out with the mechanistic probe inserted at two different sites on the reagents. If we are studying a reaction between a nucleophile and an electrophile, it may be possible to make Hammett plots from the variation of substituents on both reagents. The acylation of amines with acid chlorides is an example.



If we vary the structure of the acid chloride we get a ρ value of +1.2, typical of nucleophilic attack on the carbonyl group. If we vary the amine we get a ρ value of -3.2, typical of a reaction in which electrons that were conjugated round the ring move away to form a new bond. Comparing the numbers tells us the rate depends on the nucleophilicity of the amine 100 times more than on the electrophilicity of the acid chloride.



Non-linear Hammett plots

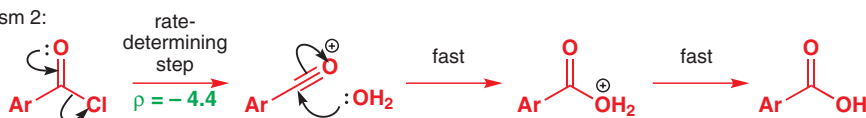


The Hammett plot for hydrolysis of the acid chlorides of benzoic acids in aqueous acetone is very odd indeed. Hammett plots need not be perfectly linear to give useful information, but this one is clearly made up of two intersecting straight lines. This might look like disaster at first but, in fact, it tells us something rather important. The right-hand part of the curve, where the more electron-withdrawing substituents lie, has a slope of +2.5: just what we should expect for rate-determining attack of water on the carbonyl group. As we go to less electron-withdrawing substituents, the rate of the reaction suddenly starts to increase as we pass the *para*-chloro compound and the left-hand part of the curve has a slope of -4.4.

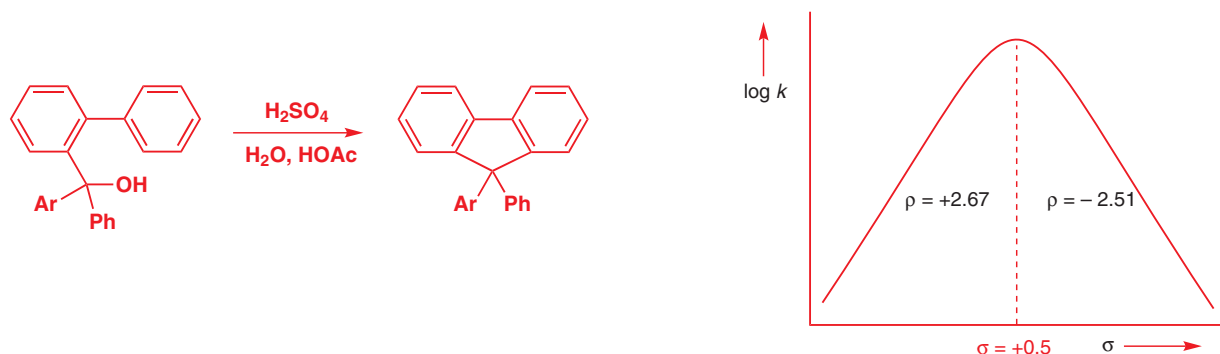
What can this mean? If the reaction becomes faster as we pass the discontinuity in the curve—and it gets faster whether we go from right to left or left to right—there must be a change in mechanism. If there is a choice between two mechanisms, the faster of the two will operate. Mechanism 1 is the rate-determining nucleophilic attack by water on the carbonyl group.

The new mechanism goes faster for more electron-donating substituents and has quite a large negative ρ value, suggesting the formation of a cation in the rate-determining step. This mechanism (mechanism 2) must surely be the S_N1 -like preliminary formation of an acylium ion by loss of chloride ion.

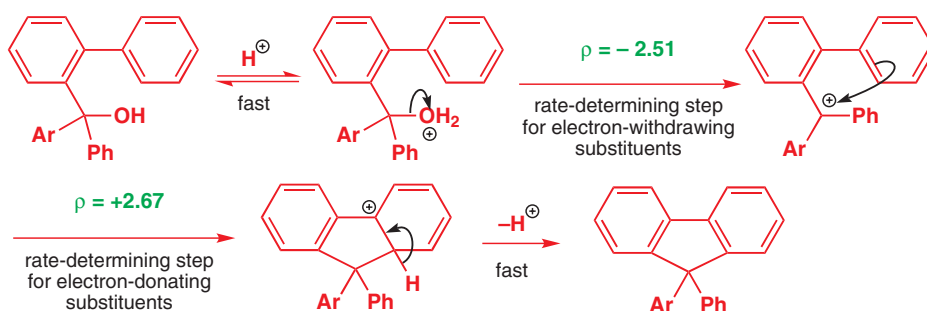
mechanism 2:



When the Hammett plot bends the other way, so that the rate of the reaction decreases as it passes the discontinuity, we have a single mechanism with a change in rate-determining step. A reaction goes by the fastest possible mechanism but its rate is limited by the slowest of the steps in that mechanism. An example is the intramolecular Friedel–Crafts alkylation of a diphenyl derivative where the alkylating agent is a diarylmethanol attached to one of the benzene rings in the *ortho* position.



The carbocation intermediate in the Friedel–Crafts reaction (Chapter 21) is rather stable, being tertiary and benzylic, and the formation of the cation, normally the rate-determining step, with inevitably a negative ρ value, goes faster and faster as the electron-donating power of the substituents increases until it is faster than the cyclization, which becomes the rate-determining step. The cyclization puts electrons back into the carbocation and has a positive ρ value. As the two steps have more or less the reverse electron flow to and from the same carbon atom, it is reasonable for the size of ρ to be about the same but of opposite sign.



- A reaction occurs by the faster of two possible mechanisms but by the slower of two possible rate-determining steps.

We shall see more examples of Hammett ρ values used in conjunction with other evidence as the chapter develops but now it is time to look at what other evidence is available.

Other kinetic evidence for reaction mechanisms

The kinetic isotope effect

■ You have seen how isotopes have different *physical* properties—their nuclear spin, for example, which affects how they behave in an NMR machine. We also showed you in Chapter 3 how IR stretching frequencies depend on mass, and there you saw that C–D bonds have lower stretching frequencies than C–H bonds. That fact is highly relevant to the explanation we are about to give you for the origin of kinetic isotope effects.

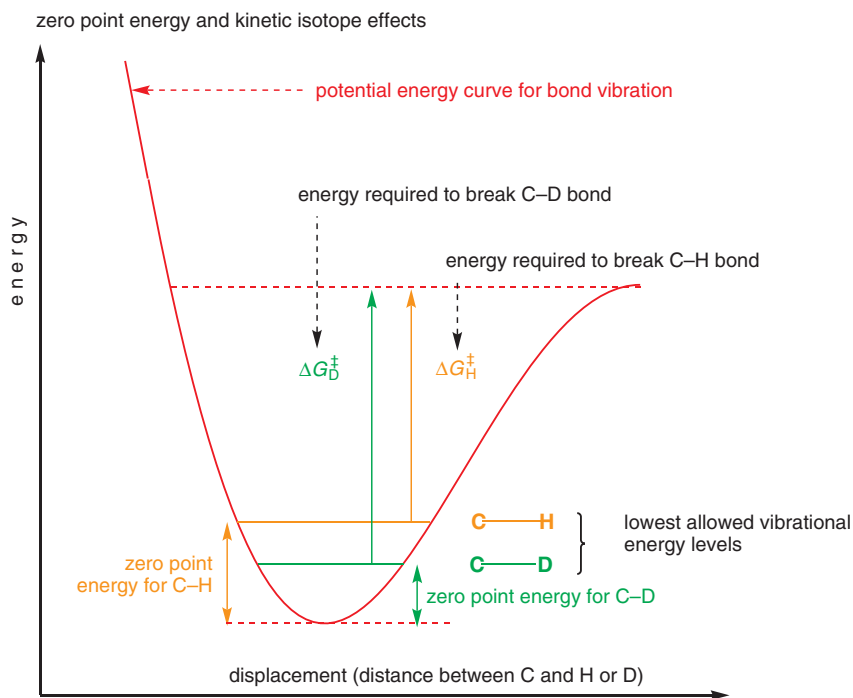
■ The full explanation of the theory of kinetic isotope effects is beyond the scope of this book but you can read about it in any book on physical organic chemistry.

Up to now you have probably (and rightly) assumed that isotopes of an element are chemically identical. They differ only in the number of neutrons in their nuclei: chemistry generally depends on charge, orbitals, and electrons. It may come as a surprise to find that this is not quite true. Isotopes may differ chemically, because some chemical properties do depend on atomic mass. However, this difference is only significant for hydrogen—no other element has one isotope twice as massive as another! *Kinetic isotope effects* are the changes in rate observed when a (^1H) hydrogen atom is replaced by a (^2H) deuterium atom in the same reaction. For any reaction, the kinetic isotope effect (KIE) is defined as

$$\text{KIE} = k_{\text{H}}/k_{\text{D}}$$

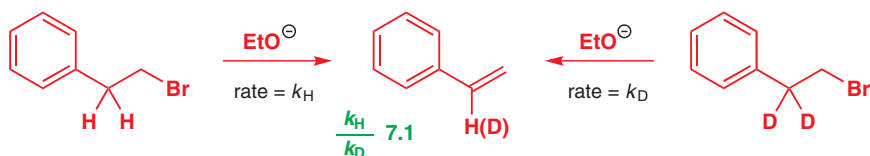
where k_{H} is the rate with a ^1H atom in the molecule and k_{D} is the rate with a ^2H (deuterium, D) atom in the molecule.

How do kinetic isotope effects come about? Even in its lowest energy state a covalent bond never stops vibrating. If it did it would violate a fundamental physical principle, Heisenberg's uncertainty principle, which states that position and momentum cannot be known exactly at the same time: a non-vibrating pair of atoms have precisely zero momentum and precisely fixed locations. The minimum vibrational energy a bond can have is called the zero point energy, and the zero point energy depends on the mass of the atoms attached to the bond—heavier atoms have a lower zero point energy than lighter ones.



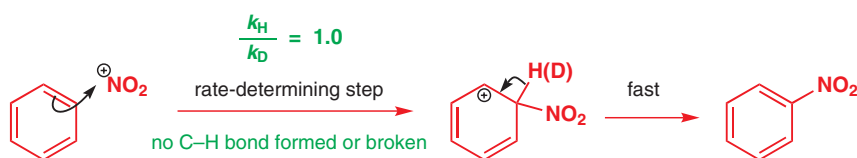
In order to break a covalent bond, a certain amount of energy is required to separate the nuclei from their starting position. This energy has to raise the vibrational state of the bond to the point where it breaks. For the sake of argument, imagine taking a C–H bond in its lowest energy state and breaking it—the diagram shows the amount of energy required, which we can call $\Delta G_{\text{H}}^{\ddagger}$. Now do the same for a C–D bond: because the zero point energy of a C–D bond is smaller than that for a C–H bond, the C–D bond needs that little bit more energy $\Delta G_{\text{D}}^{\ddagger}$ to break: in other words a C–D bond is marginally stronger than a C–H bond. This means reactions in which C–H bonds break go faster than reactions in which C–D bonds break, *providing the bond to H (or D) is involved in the rate-determining step*. The theoretical maximum value of

the KIE is about 7 for reactions at room temperature in which a bond to H or D is being broken. For example, the rates of these two eliminations can be compared, and $k_{\text{H}}/k_{\text{D}}$ turns out to be 7.1 at 25 °C.

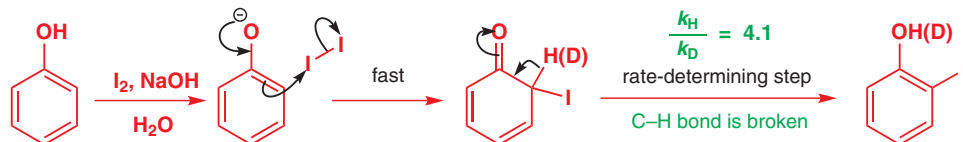


In this case the fact that the KIE is non-zero tells us that the C–H (or C–D) bond is being broken during the rate-determining step, and so the reaction must be an E2 elimination. In E1 eliminations, the rate-determining step does not involve a breaking C–H bond.

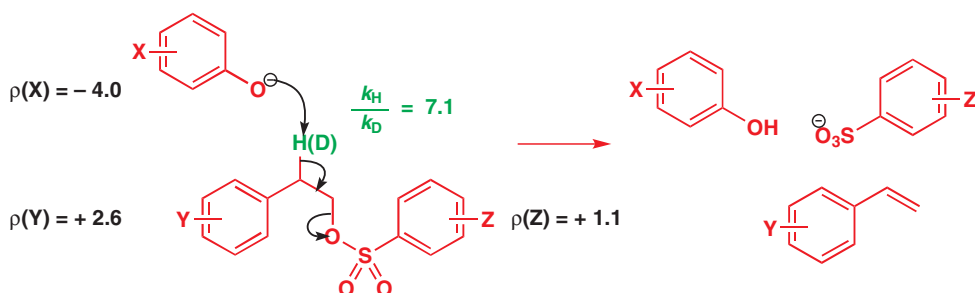
In Chapter 21 we told you that the rate-determining step in the nitration of benzene was the attack of the electrophile on the benzene ring. This is easily verified by replacing the hydrogen atoms round the benzene ring with deuteriums. The rate of the reaction stays the same, so the C–H (or C–D) bonds cannot be involved in the rate-determining step. If the second step, which does involve the breaking of a C–H bond, were the rate-determining step it would go more slowly if the H were replaced by D.



By contrast, for the iodination of phenol in basic solution there is a deuterium isotope effect of $k_{\text{H}}/k_{\text{D}} = 4.1$. Clearly, the loss of the proton from the intermediate must now be the rate-determining step—the phenolate ion reacts so rapidly that the first step is faster than the second.



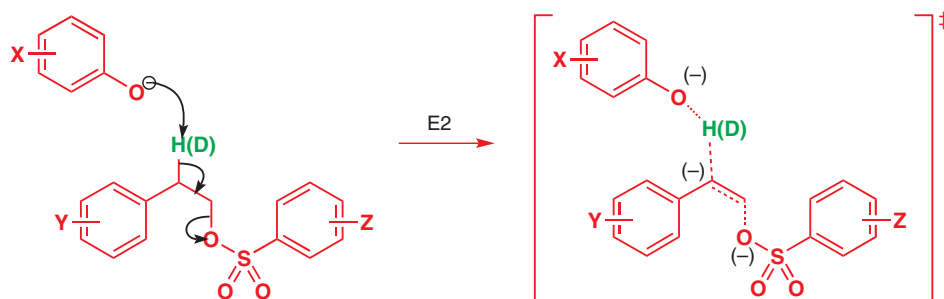
The deuterium isotope effect can add to the information from Hammett plots in building up a picture of a transition state. Three separate Hammett ρ values can be measured for the elimination reaction and this information is very valuable. In addition, a large KIE $k_{\text{H}}/k_{\text{D}} = 7.1$ is observed for the hydrogen atom under attack.



It is no surprise that the base (ArO^-) donates electrons and the leaving group (ArSO_3^-) accepts them, as the ρ values indicate. The large deuterium isotope effect tells us that the reaction is E2, but additional information comes from the moderate positive $\rho(\text{Y})$ value for the aromatic ring adjacent to the proton being lost. It might have been expected that this ring is merely a spectator, but in fact the reaction must involve a build-up of negative charge, which can be stabilized by an electron-donating substituent Y. This can be explained if we assume

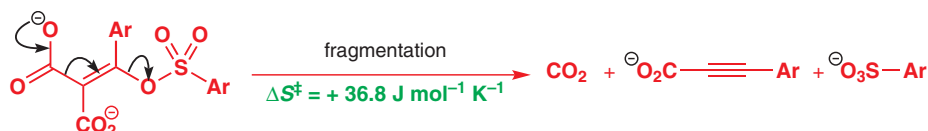
➡ E1 and E2 mechanisms were covered in Chapter 17.

that the removal of the proton is slightly more advanced at the transition state than loss of the leaving group.



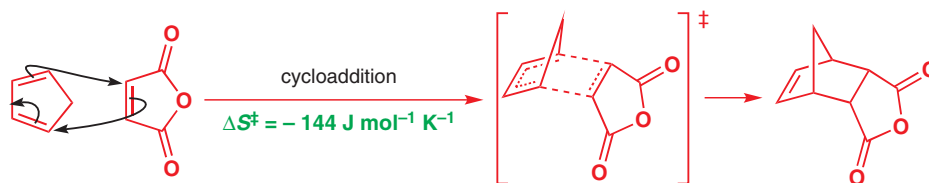
Entropy of activation

The entropy of activation, $\Delta S^\ddagger$, of a reaction tells us about the increase or decrease in order in a reaction as the starting material goes to the transition state. A positive $\Delta S^\ddagger$ means an increase in entropy or a decrease in order, and a negative $\Delta S^\ddagger$ means an increase in order. Normally, unimolecular reactions in which one molecule gives two products have a positive $\Delta S^\ddagger$ and bimolecular reactions have a negative $\Delta S^\ddagger$. Fragmentations (Chapter 36), such as this decarboxylation in which one molecule fragments to three, have positive values of $\Delta S^\ddagger$, in this case $\Delta S^\ddagger = +36.8 \text{ J mol}^{-1} \text{ K}^{-1}$.



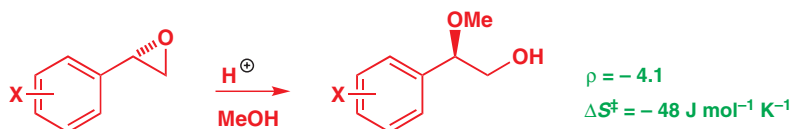
■ Entropies of activation are measured in units of $\text{J mol}^{-1} \text{ K}^{-1}$. All the values in this book are in $\text{J mol}^{-1} \text{ K}^{-1}$ but in older books you will see 'entropy units' (e.u.), which are $\text{cal mol}^{-1} \text{ K}^{-1}$. Values in e.u. should be multiplied by 4.18 to get values in $\text{J mol}^{-1} \text{ K}^{-1}$.

At the other extreme are cycloadditions (Chapter 34) such as the Diels–Alder reaction we examined a few pages back. Not only do two reagents become one product but a very precise orientation is required in the transition state, usually meaning a large negative $\Delta S^\ddagger$. Diels–Alder reactions usually have $\Delta S^\ddagger$ of about -120 to $-160 \text{ J mol}^{-1} \text{ K}^{-1}$. The classic cyclopentadiene addition to maleic anhydride has $\Delta S^\ddagger = -144 \text{ J mol}^{-1} \text{ K}^{-1}$.



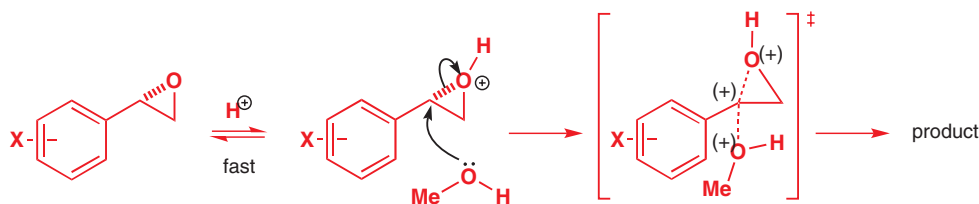
 Interactive mechanism for the Diels–Alder reaction

These numbers give you the range of entropies of activation you may expect to find. Large negative numbers are common but only small positive numbers are found. The largest negative numbers apply to bimolecular reactions where neither reagent is in great excess. Smaller negative numbers may mean a bimolecular reaction with solvent or some other reagent in large excess. The acid-catalysed opening of styrene oxides in methanol is a good example.



The Hammett ρ value of -4.1 suggests a carbocation intermediate, as does the regioselectivity of the reaction (MeOH attacks the benzylic position) but the stereochemistry (the reaction occurs with inversion) and a modest negative entropy of activation ($\Delta S^\ddagger = -48 \text{ J mol}^{-1} \text{ K}^{-1}$)

suggest rather an S_N2 reaction with a loose transition state having substantial positive charge at the benzylic carbon. Neither piece of evidence alone would be enough to define the mechanism.



Acid and base catalysis

As you have seen throughout this book, acids and bases provide the most widely used ways of speeding up reactions. If you want to make an ester—add some acid. If you want to hydrolyse an ester—add some base. We explained in Chapter 12 the ways in which acid and base catalysts help reactions along, and we introduced you to the terms **specific acid** and **specific base**, **general acid** and **general base**. We will now look in a little more detail at these types of catalysis and give some pointers as to how to establish which of them, if any, is operative in any given reaction.

As a preliminary, let's look at an example of **specific acid catalysis**. This is the kind operating in the reaction just above—epoxides don't react with methanol but, if we protonate the epoxide first, then the reaction works. Specific acid catalysis protonates electrophiles and makes them more electrophilic.

➡ For the earlier discussion of acid and base catalysis, and an outline of what the terms **specific** and **general** acid/base catalysis mean, see Chapter 12, pp. 262–264.

specific acid catalysis



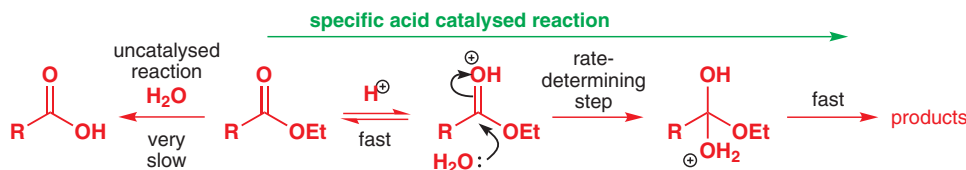
We could, on the other hand, have reasoned that although methanol is not a good enough nucleophile, deprotonating with a base will make it into the much more nucleophilic methoxide, and the reaction will also work. This sort of base catalysis—deprotonating nucleophiles to make them more nucleophilic—is **specific base catalysis**.

specific base catalysis



Specific acid catalysis

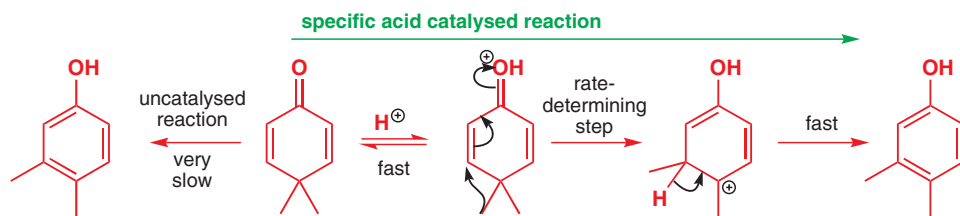
Specific acid catalysis (SAC) involves a rapid protonation of the compound followed by the slow step, which is accelerated in comparison with the uncatalysed reaction because of the greater reactivity of the protonated compound. You have just seen an example with an epoxide; ester hydrolysis (or formation) is another, as you saw in Chapter 12.



A more interesting reaction is the dienone–phenol rearrangement. Rearrangement in the absence of acid is very slow but once the ketone oxygen is protonated, it occurs very rapidly.

➡ You met this rearrangement in Chapter 36.

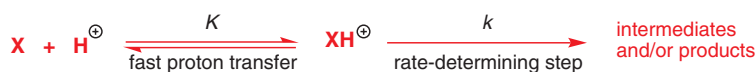
Again we have fast equilibrium protonation, followed by a rate-determining step involving a reaction of the protonated species: this is SAC.



This catalysis depends only on the protonating power of the solution. The compound must be protonated to react, so the catalyst must be a strong enough acid to do the job. It is not necessary that every molecule is protonated—just enough to set the reaction going because the catalytic acid is regenerated at the end. In a specific acid catalysed reaction, the rate of the reaction depends on the pH of the reaction mixture. SAC works only if the pH is similar to, or below, the pK_a of the conjugate acid of the substrate, and the log of the rate of the reaction is proportional to the pH of the solution.

There is one rather remarkable experimental indication of this mechanism. If the reaction is carried out in a deuterated solvent (D_2O instead of H_2O) the rate of the reaction increases. This is a *solvent isotope effect* rather than a kinetic isotope effect and needs some explanation. If you examine the three examples of SAC in the previous pages you will see that they share these characteristics: a fast proton exchange is followed by a rate-determining step that does *not* involve the making or breaking of any bonds to hydrogen. In general terms:

■ As you saw on p. 1050, the usual kinetic isotope effect has $k_H/k_D > 1$. Reactions with $k_H/k_D < 1$ exhibit an *inverse kinetic isotope effect*. We don't need to go into details of how this might come about, but this is the reason a reaction that goes faster in D_2O than in H_2O (even when that is expected) is said to have an *inverse solvent kinetic isotope effect*.



The rate of the reaction is the rate of the rate-determining step:

$$\text{rate} = k[XH^+]$$

The concentration of the intermediate $[XH^+]$ is related to the pH and to the concentration of the substrate by the equilibrium constant, K , of the protonation. This gives us:

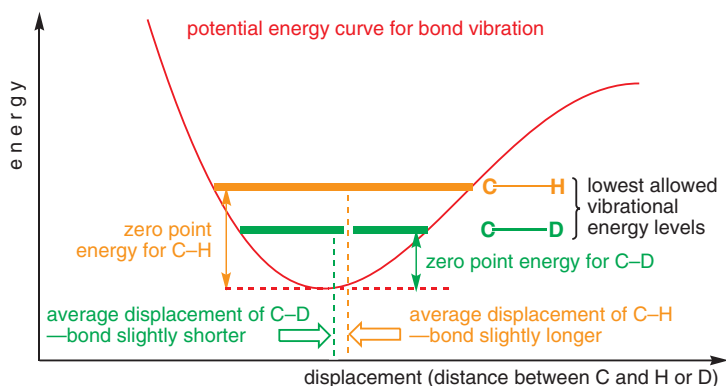
$$\text{rate} = kK[H^+][X]$$

In the acid-catalysed reaction, the bond to H (or D) is not broken in the rate-determining step, so k cannot change when hydrogen is replaced by deuterium. That means that if a reaction goes faster in D_2O than in H_2O then it must be K that is different (i.e. larger) in D_2O . SAC is more effective with D_3O^+ in D_2O than with H_3O^+ in H_2O because more of the substrate is protonated at any one time.

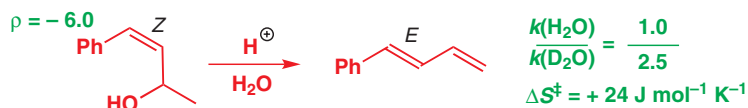
● An inverse solvent isotope effect ($k[D_2O] > k[H_2O]$) is indicative of specific acid catalysis.

■ It is not, of course, possible to use D_3O^+ in H_2O as H and D exchange very quickly. The solvent determines which acid is present.

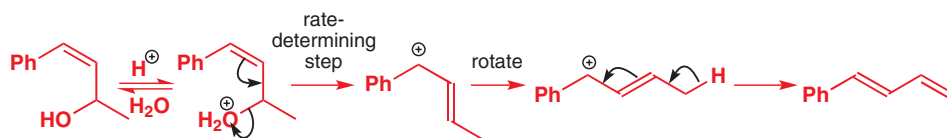
This is sometimes explained by saying that D_3O^+ is a stronger acid than H_3O^+ . This is partly true. The full truth is that D_3O^+ in D_2O is a stronger acid than H_3O^+ in H_2O . Water (H_2O) is a better solvating agent for H_3O^+ than D_2O is for D_3O^+ because O–H bonds are longer than O–D bonds. Look again at the potential energy curve we showed you on p. 1050 and reproduced below, this time representing the energies of O–H and O–D bonds. The average length of a bond is the mid-point of the line in the potential energy well representing its energy level. You can easily see that the mid-point for the O–H is further out than the mid-point for the O–D bond because of the asymmetry of the well. O–H bonds are longer than O–D bonds, and can therefore make *stronger hydrogen bonds*. These hydrogen bonds are better at allowing solvation of H_3O^+ , making H_3O^+ in H_2O less willing to protonate a substrate than D_3O^+ in D_2O .



Let's illustrate all this with an example. The *Z* allylic alcohol below dehydrates in acid solution to the *E* diene. We have lots of data on this mechanism, all summarized in the diagrams. You may like to note as well that the product contains no deuterium after dehydration in D_2O .



The Hammett ρ value of -6.0 suggests a carbocation intermediate and the positive entropy of activation suggests a rate-determining step in which disorder increases, perhaps one molecule breaking into two. The inverse solvent deuterium isotope effect (faster reaction in D_2O than in H_2O) strongly suggests SAC. Putting all this together we have a mechanism—a simple example of SAC. There is no protonation at carbon.



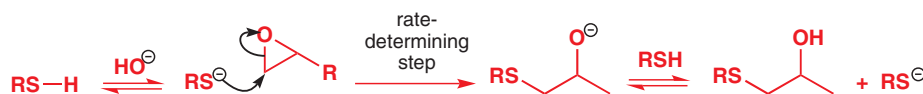
■ You can compare this mechanism with the isomerization of the same diene described earlier in this chapter.

● Summary of features of specific acid catalysis

- 1 Only H_3O^+ is an effective catalyst; pH alone matters.
- 2 Usually means rate-determining reaction of protonated species.
- 3 Effective only at pHs near or below the pK_a of the substrate's conjugate acid.
- 4 Proton transfer is not involved in the rate-determining step.
- 5 Only simple unimolecular and bimolecular steps—moderate + or $-\Delta S^\ddagger$.
- 6 Inverse solvent isotope effect $k(H_2O) < k(D_2O)$.

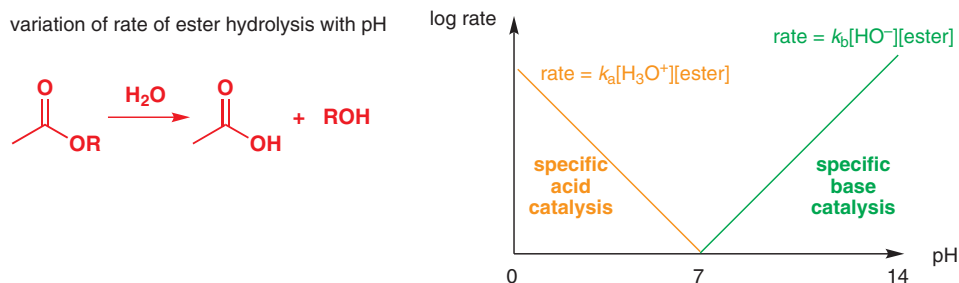
Specific base catalysis

The other side of the coin is *specific base catalysis* (SBC). SBC usually involves the removal of a proton from the substrate in a fast pre-equilibrium step followed by a rate-determining reaction of the anion. Most of the base-catalysed reactions you are familiar with work by SBC. Examples include opening of epoxides with thiols.

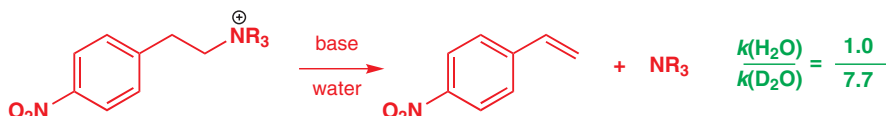


As with SAC, the rate of the reaction depends on the pH of the solution. If it is around or higher than the pK_a of the thiol, thiolate anion will be formed and this opens the epoxide much faster than does the unionized thiol. The nucleophile is then regenerated by the oxyanion produced in the rate-determining step.

It is quite common for specific acid and specific base catalysis to operate on the same reaction, depending on the pH at which the reaction is carried out. In fact, you have already seen this for ester hydrolysis in Chapter 12. The pH–rate profile (Chapter 12) for the hydrolysis of a simple ester such as ethyl acetate shows just two straight lines meeting each other (and zero rate) at about neutrality. Ethyl acetate hydrolysis occurs by SAC or SBC only.

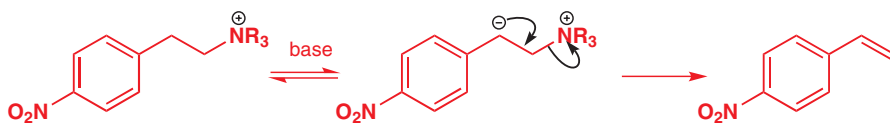


Removal of a proton from heteroatoms by heteroatom bases is never rate determining because it is always fast, but removal of a proton from carbon *can* be the rate-determining step. A remarkably large inverse solvent deuterium isotope effect was found with this elimination of a tertiary amine in basic solution.



➡ E1, E2, and E1cB mechanisms are described in Chapter 17.

The detailed mechanism cannot be E2 or the isotope effect, if any, would be the other way round. With SBC, however, the mechanism can be E1cB having a carbanion as an intermediate.



Microscopic reversibility

There is only one least-energy pathway between two interconverting compounds such as the starting material and the intermediate here. Every microscopic detail of the back reaction is exactly the same as that for the forward reaction. This is the principle of microscopic reversibility. Here we use evidence from the back reaction (slow proton transfer from water to the carbanion) to tell us about the forward reaction.

The isotope effect observed is certainly inverse (the reaction is faster with H_2O than D_2O) but the magnitude of the effect is too large to be a *solvent* isotope effect and looks much more like an inverse *kinetic* isotope effect. And so it is. The tertiary amine is not a very good leaving group in spite of its positive charge (pK_a of R_3NH^+ is about 10) so the carbanion mostly reverts to starting materials. The isotope effect is a kinetic isotope effect on this reverse step—the protonation of the carbanion. This reaction involves a proton transfer from H_2O or D_2O and will be much faster (7.7 times in fact) in H_2O due to an ordinary kinetic isotope effect. The *elimination* reaction goes faster in D_2O because the back reaction goes more slowly and more of the carbanion goes on to product.

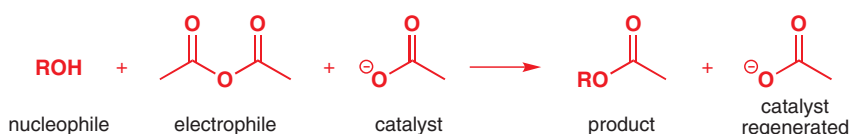
• Summary of features of specific base catalysis

- 1 Only HO^- is an effective catalyst; pH alone matters.
- 2 Usually means rate-determining reaction of deprotonated species.
- 3 Effective only at pHs near or above the pK_a of the substrate.
- 4 Proton transfer is not involved in the rate-determining step, unless C–H bonds are involved.
- 5 Only simple unimolecular and bimolecular steps—moderate + or $-\Delta S^\ddagger$.
- 6 Inverse solvent isotope effect $k(H_2O) < k(D_2O)$.

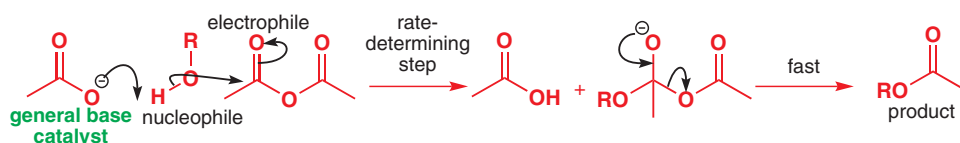
General base catalysis

In Chapter 12 (p. 263) we pointed out that even weak bases—too weak to deprotonate a nucleophile by the mechanism we have just described for SBC—can still act as catalysts. Such catalysts are known as **general base catalysts**, and are the promoters of a parallel kind of acid–base catalysis called ‘general’ rather than ‘specific’. General base catalysis, abbreviated GBC, depends not only on pH (i.e. the concentration of hydroxide ion) but also on the concentration of other bases too. General acid catalysis, abbreviated GAC, likewise depends not only on pH (i.e. the concentration of H_3O^+) but also on the concentration of other undissociated acids HA. General acid–base catalysis is a milder kind of catalysis and is characteristic of reactions catalysed by enzymes in the metabolism of living things.

In a general base-catalysed reaction, proton transfer is not complete before the rate-determining step (as it was in SBC) but occurs *during* the rate-determining step. A simple example is the catalysis by acetate ion of the formation of esters from alcohols and acetic anhydride.

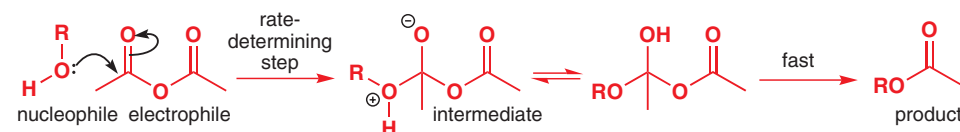


How can this catalysis work? At first sight there seems to be no mechanism available. Acetate cannot act as a specific base—it is far too weak ($\text{p}K_{\text{a}} \text{ AcOH } 4.7$) to remove a proton from an alcohol ($\text{p}K_{\text{a}}$ about 15). It can't operate as a nucleophile, as pyridine does (p. 200), as nucleophilic attack on acetic anhydride would be a non-reaction, simply regenerating starting materials. The only thing it can do is to remove the proton from the alcohol *as the reaction occurs*.



You will see at once that there is a great disadvantage in this mechanism: the rate-determining step is termolecular—three molecules have to collide. This comes out most clearly in the entropy of activation, which has an enormous negative value—around $\Delta S^\ddagger = -168 \text{ J mol}^{-1} \text{ K}^{-1}$ for this reaction. For this reason, GBC or GAC reactions are normally effective only if one of the three molecules is present in large excess—this reaction might be done in ROH as a solvent, for example, so that ROH is always present. We would also expect a normal kinetic isotope effect for ROD compared with ROH as a bond to hydrogen is being formed and broken in the rate-determining step: it is $k_{\text{H}}/k_{\text{D}} = 2.4$ here.

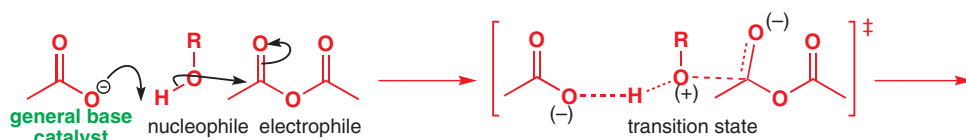
In understanding how this GBC works it is helpful to look at the mechanism without catalysis.



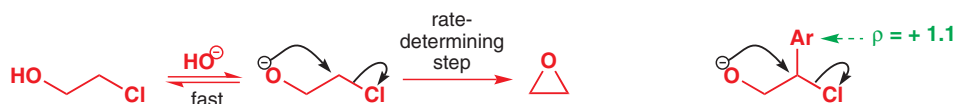
The acetate catalyst cannot remove a proton from the starting material but it can easily remove a proton from the intermediate, which has a complete positive charge on the alcohol oxygen atom. The starting material has a $\text{p}K_{\text{a}}$ above the $\text{p}K_{\text{a}}$ of HOAc but the product has a $\text{p}K_{\text{a}}$ well below it. Somewhere in the middle of the rate-determining step, the $\text{p}K_{\text{a}}$ of the ROH proton passes through the $\text{p}K_{\text{a}}$ of acetic acid and then acetate is a strong enough base to remove it. The GBC is effectively deprotonating the transition state.

➡ There was some discussion of this reaction in Chapter 12. Chapter 10 refers to the difficulty of pinpointing proton transfers in mechanisms involving the carbonyl group.

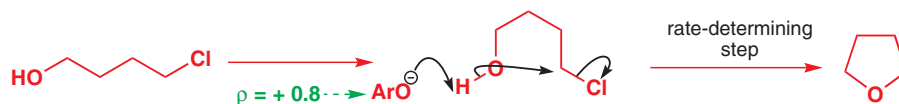
➡ The first time you met third-order kinetics (where it arose from a combination of more than one step, see p. 261) we pointed out how unlikely real termolecular steps are.



So how do we find GAC or GBC? Well, first we must remove the more powerful ‘specific’ style of catalysis by working at constant pH because SAC or SBC depends on pH alone. If we find that the rate of the reaction changes with the concentration of a weak base at constant pH, we have GBC. The formation of three- and five-membered cyclic ethers shows the contrast between GBC and SBC. The formation of epoxides is straightforward SBC with a simple linear dependence on pH between pH 8 and 12, and no acceleration at constant pH by carbonate (CO_3^{2-}) ions. There is an inverse solvent isotope effect and an aryl substituent at the electrophilic carbon atom gives the small positive ρ value expected for $\text{S}_{\text{N}}2$ with an anion.



Formation of tetrahydrofuran (THF) is also faster at higher pH but, by contrast, is additionally accelerated by various bases at constant pH. If anions of phenols (ArO^-) are used as catalysts, a Hammett ρ value of +0.8 shows that electrons are flowing away from the aromatic ring. There is a small normal kinetic isotope effect $k_{\text{H}}/k_{\text{D}} = 1.4$. Both SBC and GBC are therefore operating in this reaction. Here is the mechanism with ArO^- as GBC.



Why are the two different? The THF is easy to form, the transition state is unstrained, and only a little help is needed to make the reaction go—GBC will do. The epoxide is very strained indeed and the starting material needs to be raised in energy before cyclization will occur. Only the most powerful form of catalysis—SBC—is good enough.

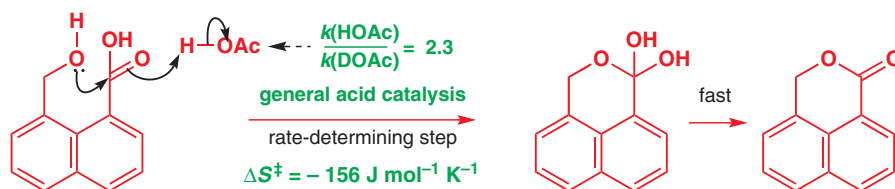
• Summary of features of general base catalysis

- 1 Any base is an effective catalyst; pH also matters.
- 2 Proton transfer is involved in the rate-determining step.
- 3 Effective at neutral pHs even if below the $\text{p}K_{\text{a}}$ of the substrate.
- 4 Catalyst often much too weak a base to deprotonate the reagent.
- 5 Catalyst removes a proton, which is becoming more acidic in the rate-determining step.
- 6 Some other bond making or bond breaking also involved unless proton is on carbon.
- 7 Often termolecular rate-determining step: large $-\Delta S^\ddagger$.
- 8 Normal kinetic isotope effect $k(\text{H}) > k(\text{D})$.

General acid catalysis

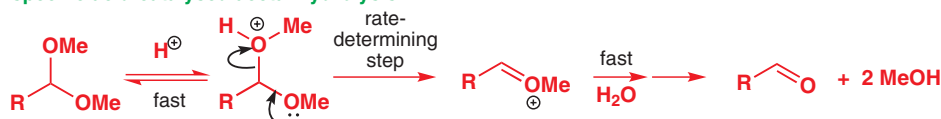
GAC involves transfer of a proton from a weak acid (too weak to protonate the substrate completely) *during* the rate-determining step. A few examples will demonstrate to you how this works. They are all examples where GAC occurs because of a modification to a familiar reaction involving SAC.

In the first one, the termolecular problem (i.e. the fact that in GAC and GBC three molecules have to come together in the transition state) is avoided by making a reaction intramolecular. Normally, ester formation and hydrolysis are specific-acid-catalysed only, but here there is catalysis by a weak acid: acetic acid. A normal kinetic isotope effect $k(\text{HOAc})/k(\text{DOAc}) = 2.3$ shows that proton transfer occurs in the rate-determining step and there is a large negative $\Delta S^\ddagger = -156 \text{ J mol}^{-1} \text{ K}^{-1}$. This is GAC of nucleophilic attack on a carbonyl group, admittedly in a rather special molecule.

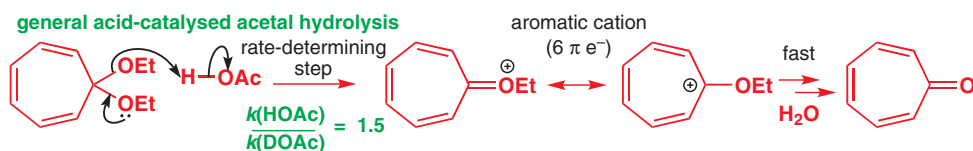


In Chapter 11 we emphasized the importance of the mechanism for the formation and hydrolysis of acetals. These are SAC reactions: alcohols are bad leaving groups and usually need to be fully protonated by strong acids before they will go, even with the help of the lone pair of another oxygen atom.

specific acid-catalysed acetal hydrolysis

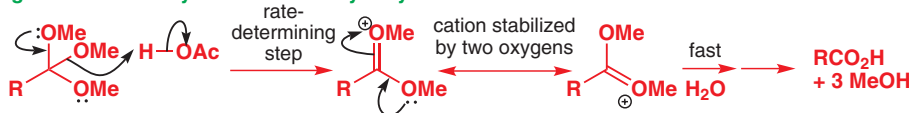


If we speed up the slow step by adding to the molecule some feature that stabilizes the cation intermediate, GAC may be found. One example is the aromatic cation formed in the hydrolysis of cycloheptatrienone acetals. The normal kinetic isotope effect announces the appearance of GAC.



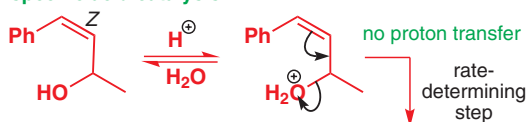
Even adding one extra alkoxy group so that we have an orthoester instead of an acetal is enough. These compounds show catalysis with a variety of weak acids at not very acidic pH (5–6). As one OMe group is protonated, two others help in pushing it out, and they both help to stabilize the intermediate cation. Nature prefers these milder methods of catalysis, as we will see in Chapter 42.

general acid-catalysed orthoester hydrolysis



For another contrast between SAC and GAC we need only refer you back to the two *Z/E* isomerizations earlier in the chapter. Isomerization of the diene is GAC—protonation at carbon is the slow step—and isomerization of the allylic alcohol is SAC. What we didn't tell you earlier was that the GAC reaction has a normal kinetic isotope effect of $k(\text{H})/k(\text{D}) = 2.5$ and a negative entropy of activation $\Delta S^\ddagger = -36 \text{ J mol}^{-1} \text{ K}^{-1}$ —just what we should expect for a bimolecular reaction involving rate-determining proton transfer from oxygen to carbon. Notice that the intermediate cation is the same whichever the route; only the ways of getting there, including the rate-determining steps, are different.

specific acid catalysis



general acid catalysis



Nature often makes use of GAC and GBC since the catalysts required are compatible with the need to work at pH close to neutral. As in this example, enzymes manage to reduce the number of molecules required for GAC and GBC to operate by building the catalytic functional groups into their active sites.

➡ In these examples the steps after the rate-determining step are omitted and you should look at Chapter 11 for the full details.

These examples show you that GAC is possible with strong acids, especially when protonation is at carbon and that in such cases no other bond-making or -breaking steps need be involved.

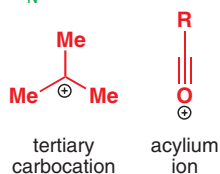
● **Summary of features of general acid catalysis**

- 1 Any acid is an effective catalyst; pH also matters.
- 2 Proton transfer is involved in the rate-determining step.
- 3 Effective at neutral pHs even if above the pK_a of the conjugate acid of the substrate.
- 4 Catalyst often much too weak an acid to protonate reagent.
- 5 Catalyst adds proton to a site that is becoming more basic in the rate-determining step.
- 6 Some other bond-making or bond-breaking also involved unless proton is on carbon.
- 7 Often termolecular rate-determining step: large $-\Delta S^\ddagger$.
- 8 Normal kinetic isotope effect $k(H) > k(D)$.

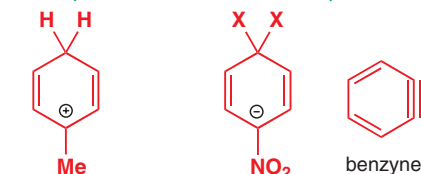
The detection of intermediates

In earlier chapters we revealed how some reactive intermediates can be prepared, usually under special conditions rather different from those of the reaction under study, as a reassurance that some of these unlikely looking species can have real existence. Intermediates of this kind include the carbocation in the S_N1 reaction (Chapter 15), the cations and anions in electrophilic (Chapter 21) and nucleophilic (Chapter 22) aromatic substitutions, and the enols and enolates in various reactions of carbonyl compounds (Chapters 20, 25, and 26). We have also used labelling in this chapter to show that symmetrical intermediates are probably involved in, for example, nucleophilic aromatic substitution with a benzyne intermediate (Chapter 22).

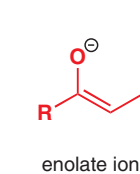
intermediate in
 S_N1 reactions



intermediates in aromatic substitution reactions
electrophilic nucleophilic



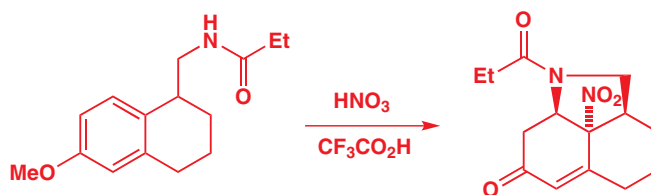
intermediate in
carbonyl reactions



We have hedged this evidence around with caution since the fact that an intermediate can be prepared does not by any means prove that it is involved in a reaction mechanism. In this section we are going to consider other and better evidence for intermediates and at the same time revise some of the earlier material.

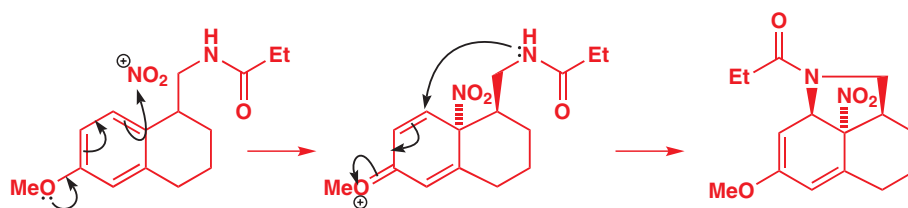
Trapping reactions

A more impressive piece of evidence is the design of a molecule that has built into it a functional group that could react with the intermediate in a predictable way but could not reasonably react with other species that might be present. For example, aromatic ethers react with nitrating agents in the *ortho* or *para* positions (Chapter 21). The intermediate has a positive charge delocalized over three of the carbon atoms in the benzene ring. If a nucleophilic group is built into the structure in the right way, it might trap this intermediate and stop it reacting further.



If we try drawing a mechanism for the formation of this remarkable compound, we discover that a necessary intermediate is also an intermediate in our preferred mechanism for aromatic nitration. The amide has trapped the cation we would propose as an intermediate in aromatic

nitration, so we feel more confident about that mechanism. The product is an enol ether that will hydrolyse to the observed enone.



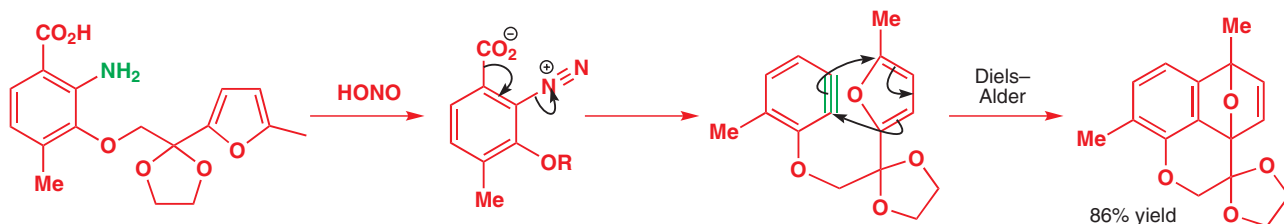
This mechanism explains everything, including the stereochemistry. The NO_2^+ attacks the aromatic ring *para* to the OMe group and on the opposite side to the amide. The amide is now in the perfect position to capture the cation at the *meta* position and, because the tether is short, it must form a *cis* bridge.

To be convincing, evidence for an intermediate should include:

- detection of the intermediate in the reaction mixture, perhaps by a trapping reaction
- a demonstration that the intermediate gives the product when added to the reaction mixture (this also means that it must be prepared as an at least reasonably stable compound)
- kinetic evidence that the rate of formation and rate of disappearance are adequate
- other suitable evidence of the kind that we have been discussing in this chapter.

A neat intramolecular trap for a benzyne works in this way. A standard benzyne-generating reaction, the diazotization of an *ortho*-amino benzoic acid (Chapter 22), gives a zwitterion that loses nitrogen and CO_2 to release the benzyne. A furan tethered to the next *ortho* position traps the benzyne in an intramolecular Diels–Alder reaction. The yield is impressive and the trap is very efficient.

➡ Why the cyclic acetal? It makes the cyclization more efficient by the Thorpe–Ingold effect (see Chapter 31).

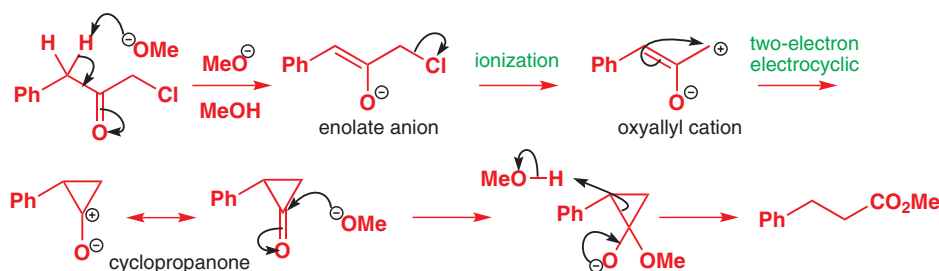


This reaction cannot really be explained without a benzyne intermediate. This same method of making benzyne is used on other *o*-amino benzoic acids and so we deduce that they presumably create benzyne too.

A collection of reactions linked by a common intermediate

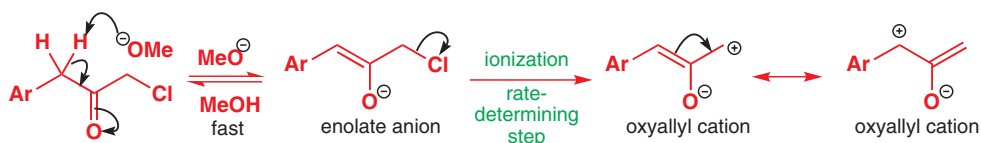
Particularly convincing evidence can develop when a number of chemists suggest the same intermediate for a number of different reactions and show that it is possible to trap the intermediate from one reaction, put it into the others, and get the normal products. We are going to describe one such set of related reactions. In Chapter 36 we suggested a mechanism for the Favorskii rearrangement involving a series of remarkable intermediates. Here is an example.

➡ The Favorskii rearrangement is on p. 950.

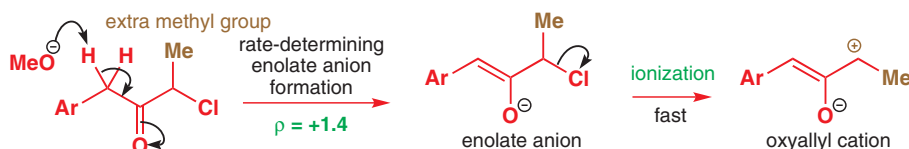
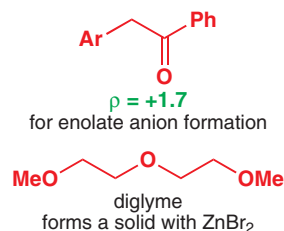


🔗 Interactive mechanism for the Favorskii rearrangement

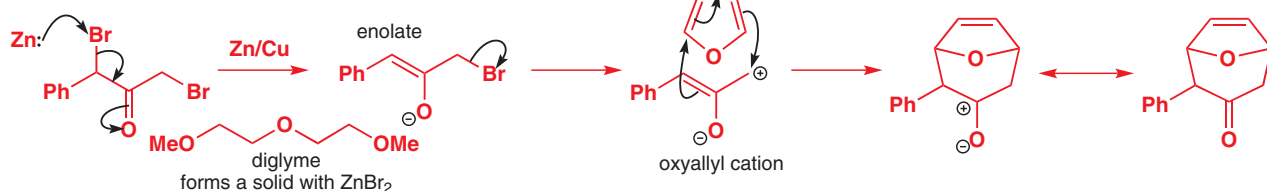
We'll summarize the evidence on this particular example. If the reaction is run in MeOD instead of MeOH, the starting material becomes deuterated at the site of enolate formation, suggesting that this is a fast and reversible step. The entropy of activation for the reaction is $\Delta S^\ddagger = +64 \text{ J mol}^{-1} \text{ K}^{-1}$, suggesting that the slow step is one molecule breaking into two. There is only one such step—the second, ionization step. If various substituted phenyl groups are used, the Hammett ρ value is -5 . This large negative value also suggests that the ionization is the slow step as the cation is delocalized into the benzene ring.



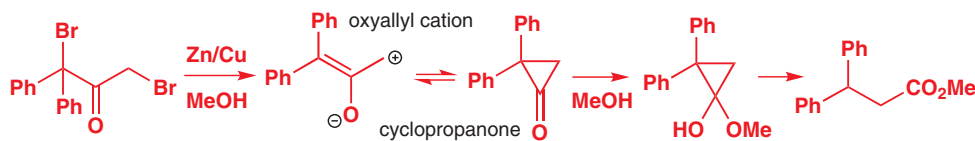
There is evidence for the first intermediate—the exchange of deuterium from the solvent. In fact formation of the enolate can even become the rate-determining step. If we merely add an extra methyl group to the chloroketone the reaction becomes 220 times faster and the rate-determining step changes. There is no longer any exchange of deuterium from the solvent and the Hammett ρ value changes from -5 to $+1.4$. This small positive value, showing some modest increase in electron density near the ring, matches typical known ρ values for enolate formation.



However, it's not too surprising that an enolate ion is formed from a ketone in basic solution. The oxyallyl cation is a much more unusual species. How can we be convinced that it really is an intermediate? One way is to make it by an alternative route. If basic nucleophiles such as the methoxide ion are avoided and reaction of zinc with an α,α' -dibromoketone in a non-nucleophilic solvent like diglyme is used instead, the oxyallyl cation can be trapped in a Diels–Alder reaction. This is the basis for a good synthesis of seven-membered rings.



But does the oxyallyl cation go on to give cyclopropanones? In fact, there is good evidence that the two are in equilibrium. If the same method is used to create the diphenyl oxyallyl cation in methanol instead of in diglyme, the normal Favorskii product is produced. Evidently, methoxide is needed only to produce the enolate—methanol alone is enough to decompose the cyclopropanone.



Further information comes from another reaction. If a suitable (1,3-di-*t*-butyl) allene is epoxidized with *m*-CPBA the unstable allene oxide can actually be isolated. On heating, this

➡ For related reactions, see Chapter 34, p. 894.

The reaction mechanism for the epoxidation of 2,3-dimethyl-1,3-butadiene with *m*-CPBA is shown. The starting material is 2,3-dimethyl-1,3-butadiene. It reacts with *m*-CPBA to form a cyclic intermediate, a three-membered ring with an oxygen atom and a double bond. This intermediate then undergoes a rearrangement step at 100 °C to form the final product, 2,3-dimethyl-oxirane (2,3-dimethyl-epoxide).

Reaction scheme showing the synthesis of 1,1-dimethyl-2,2-diphenylcyclopropanone:

Starting materials: 2-bromo-3,3-dimethylbutan-2-one and a potassium enolate of 1-chloro-4-(trimethylsilyl)benzene.

Intermediate: A potassium enolate of 2-bromo-3,3-dimethylbutan-2-one.

Product: 1,1-dimethyl-2,2-diphenylcyclopropanone.

Reaction scheme showing the synthesis of methyl tert-butyl carbonate (CO₂Me) from isobutylene oxide and diazomethane (CH₂N₂) in dichloromethane (CH₂Cl₂) at -78 °C, followed by reaction with methoxide (MeO⁻) and methanol (MeOH).

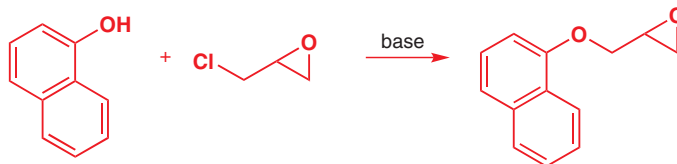
Reaction scheme showing the synthesis of a macrocyclic ketone:

- 1,1-dimethylcyclopropanone reacts with 2-methylpropanal to form a 1,5-dicarbonyl intermediate.
- The intermediate reacts with furan to form a bicyclic intermediate.
- The bicyclic intermediate undergoes ring opening to form a macrocyclic ketone.

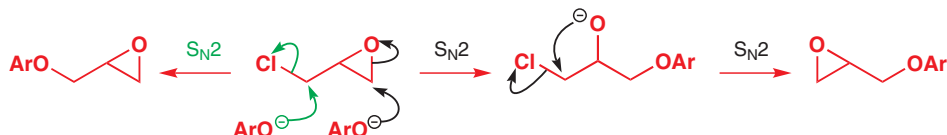
Stereochemistry and mechanism

Although we have left stereochemistry to the last, it is one of the most important tools in unravelling complex mechanisms. You have already seen how inversion of configuration is a vital piece of evidence for an S_N2 mechanism (Chapter 15) while retention of configuration is the best evidence for participation (Chapter 36). You have seen the array of stereochemical evidence for pericyclic mechanisms (Chapters 34 and 35). The chapters devoted to diastereoselectivity (32 and 33) give many examples where information about the mechanism follows from the stereochemistry. We shall not go over that material again, but summarize the types of evidence with new examples. The first example looks too trivial to mention.

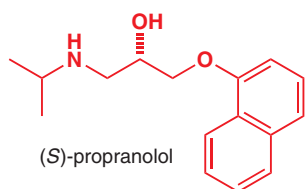
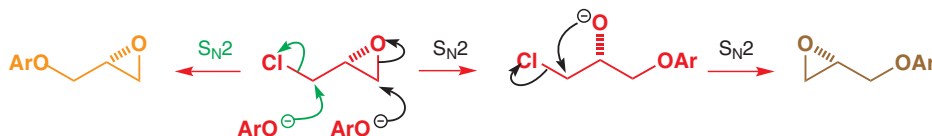
Retention or inversion?



Although this reaction looks like a simple S_N2 displacement by the naphthyloxy anion on the primary alkyl chloride, there is, in fact, a reasonable alternative—the opening of the epoxide at the less hindered primary centre followed by closure of the epoxide the other way round. The electrophile is known as epichlorohydrin and has two reasonable sites for nucleophilic attack.

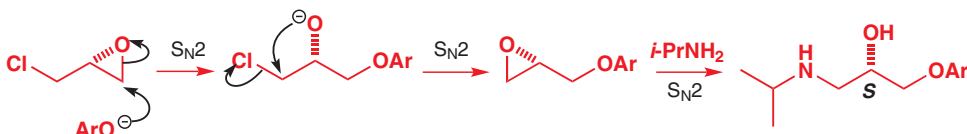


It looks difficult to tell these mechanisms apart since both involve the same kind of reaction. Stereochemistry is the answer. If enantiomerically pure epichlorohydrin is used, the two mechanisms give different enantiomers of the product. Although each S_N2 reaction takes place at a primary centre and the stereogenic centre remains the same, the products shown in orange and in brown are obviously enantiomers.

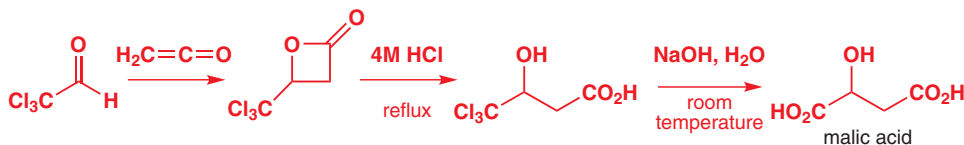


➡ A synthesis of propranolol is given in Chapter 28.

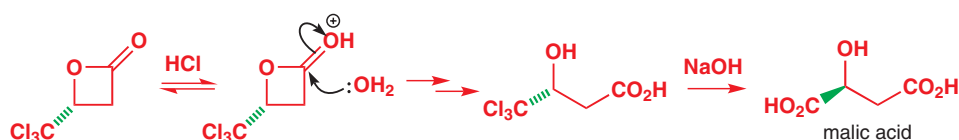
Finding out the mechanism of this process is not idle curiosity as a group of drugs used to combat high blood pressure and heart disease, such as propranolol, are made from epichlorohydrin and it is essential to know which enantiomer to use to get the right enantiomer of the drug. In fact, the epoxide is attacked initially, in preference to the chloride.



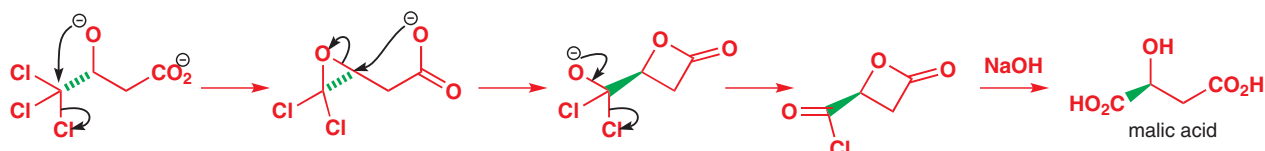
A more complicated example arises from the strange reactions used to make malic acid from chloral and ketene. An initial [2 + 2] cycloaddition is followed by acid treatment and then treatment with an excess of aqueous NaOH. Neutralization gives malic acid.



The mechanism of this reaction also looks straightforward: normal ester hydrolysis followed by hydrolysis of the CCl_3 group to CO_2H . Caution suggests investigation, particularly as four-membered lactones sometimes hydrolyse by S_N2 displacement at the saturated ester carbon rather than by attack on the carbonyl group, like the three-membered lactones discussed in Chapter 36 (p. 934). The solution was urgently needed when it was found that enantiomerically pure lactone could be prepared as a single enantiomer. The sequence was repeated with enantiomerically pure lactone: lactone hydrolysis occurred with retention of configuration and must be normal ester hydrolysis by attack of water at the carbonyl group. But the hydrolysis of the CCl_3 group surprisingly occurred with inversion of configuration.



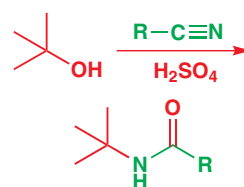
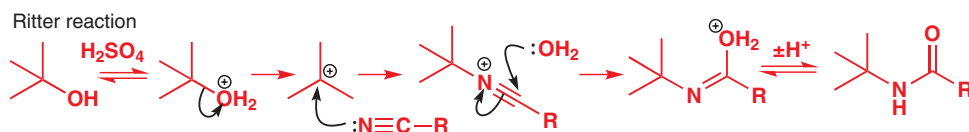
The answer must be a mechanism related to the one we have just seen for epichlorohydrin. Attack by hydroxide on CCl_3 is almost unknown and it is much more likely that intramolecular attack by alkoxide to give an epoxide should occur. The carboxylate anion can then invert the stereogenic centre by intramolecular $\text{S}_{\text{N}}2$ displacement. Notice that the tether ensures attack at the nearer end of the epoxide. The second four-membered lactone also hydrolyses by attack at the carbonyl group.



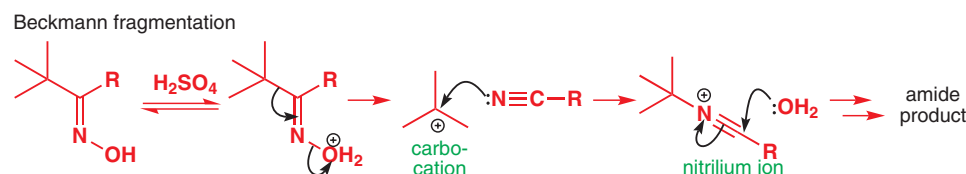
➡ In Chapter 31 we discussed Baldwin's rules for ring closures such as these. The unreactivity of the trichloromethyl group is related to the unreactivity of dichloromethane: see p. 804.

The Ritter reaction and the Beckmann fragmentation

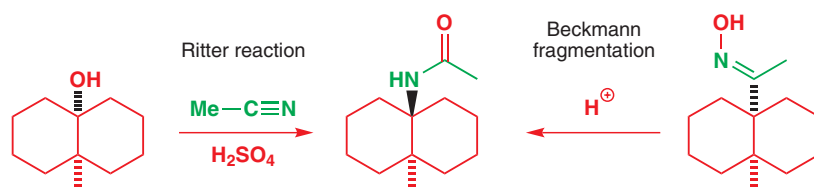
Another collection of related intermediates occurs in the Ritter reaction and the Beckmann fragmentation. The Ritter reaction involves the combination of a tertiary alcohol and a nitrile in acid solution and the proposed mechanism involves a series of intermediates.



The Beckmann fragmentation also occurs in acid solution on the fragmentation of an oxime with a tertiary alkyl group *anti* to the OH of the oxime. The fragmentation step gives the same cation and the same nitrile together with a molecule of water and these three combine in the same way to give the same amide. We need evidence that the carbocation and the nitrilium ion are genuine intermediates and that the same sequence is found in both reactions.



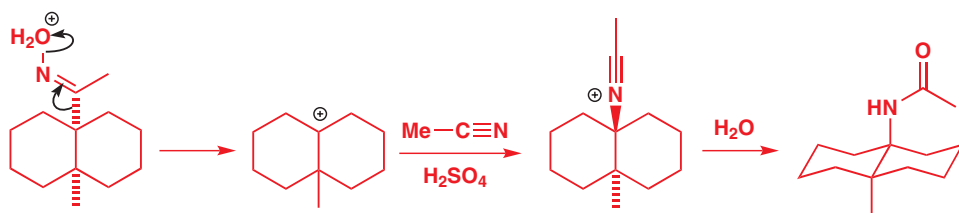
Evidence that the two reactions are intimately related comes from the formation of the same amide from two different starting materials: a tertiary alcohol and an oxime, both based on the decalin skeleton. The oxime has its OH group *anti* to the ring junction to minimize steric hindrance as oxime formation is under thermodynamic control (Chapter 11).



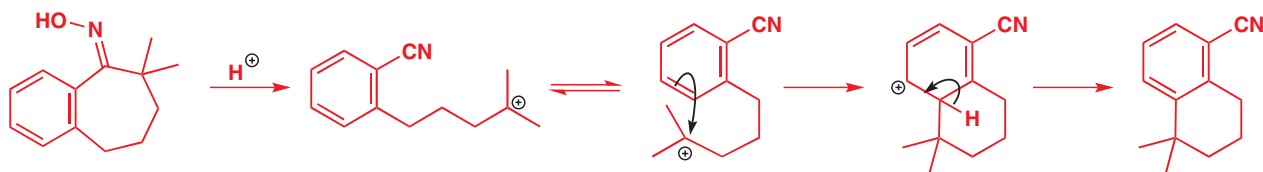
The experiments also provide stereochemical evidence that a carbocation is an intermediate in both reactions. Both starting materials are *cis*-decalins but the product is a *trans*-decalin. The carbocation intermediate has no stereochemistry and can react with the nitrile from either face: since axial attack is preferred the product is the more stable *trans*-decalin. Here's the mechanism for the Beckmann fragmentation:

➡ The Ritter reaction was introduced in Chapter 15 and the Beckmann fragmentation in Chapter 36.

 Interactive mechanism for the Beckmann fragmentation

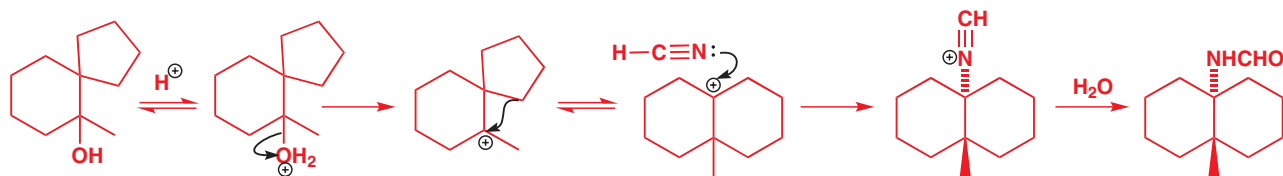


It is also possible to trap the carbocation in other ways. The Beckmann fragmentation of this oxime of an aryl seven-membered ring ketone gives a tertiary carbocation that might be expected to cyclize to give an amide. However, this reaction would give an unfavourable eight-membered ring (see Chapter 32) and does not happen. Instead, the chain twists round the other way and forms a much more stable six-membered ring by intramolecular Friedel-Crafts alkylation.



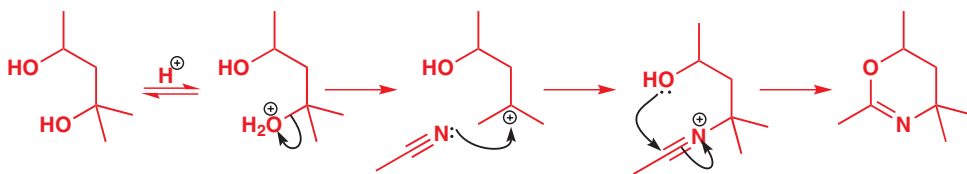
➡ Rearrangements of carbocations were described in Chapter 36, pp. 940–947.

In the Ritter reaction a rather different kind of evidence for the cation is the fact that families of isomeric alcohols all give the same product. In all these cases, rearrangements of the first-formed carbocation can easily account for the products. An example in the decalin series is this Ritter reaction with KCN as the nitrile in acidic solution so that HCN is the reagent. The starting material is a spirocyclic tertiary alcohol but the product is a *trans*-decalin formed by rearrangement.

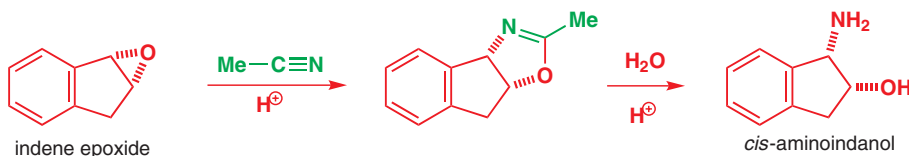


■ It's worth pointing out here that use of cyanides in acid is extremely dangerous.

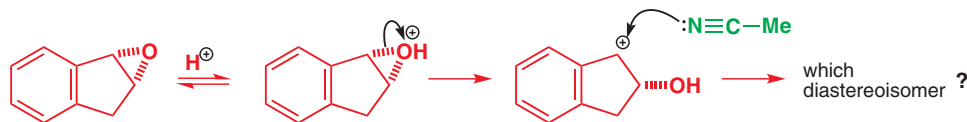
Trapping the nitrilium cation is also possible. A famous example is the heterocycle (an oxazine, Chapter 32) produced by intramolecular capture of the nitrilium ion with a hydroxyl group. Note that the tertiary alcohol reacts to give the cation while the secondary alcohol acts as the nucleophilic trap.



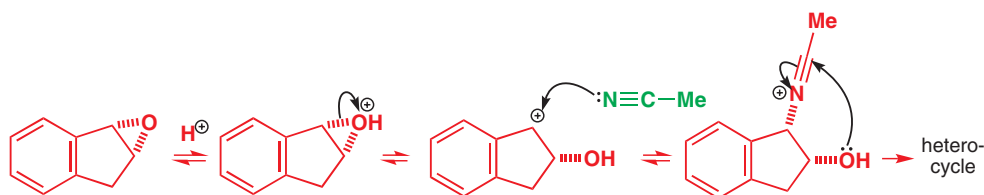
An important example in which the diastereoisomer produced was critical in determining the mechanism is the synthesis of *cis*-aminoindanol, a part of Merck's anti-HIV drug Crixivan (indinavir). The reaction involves treatment of indene epoxide with acetonitrile (MeCN) in acidic solution. The product is a *cis* fused heterocycle. It is easy to see which atoms have come from the nitrile (green) but the substitution of nitrogen for oxygen at one end of the epoxide has occurred with retention of configuration as the *cis*-epoxide has given the *cis* product. Clearly, we have some sort of Ritter reaction and the nitrilium ion has been trapped with an OH group.



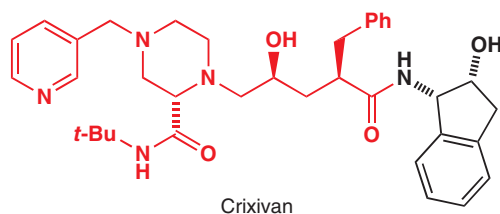
What about the regioselectivity? The obvious explanation is that a cation is formed from the epoxide by a specific acid-catalysed ring opening. But why should the nitrile attack the bottom face of the cation? We should expect it to attack the top face preferentially as the hydroxyl group partly blocks the bottom face.



A reasonable suggestion is that the nitrile adds *reversibly* to the cation. Every time it adds to the top face, it drops off again as the OH group cannot reach round to form the heterocycle. But every time it adds to the bottom face (which may well be less often), it is quickly captured by the OH group because 5,5 fused rings are favourable when the ring junction is *cis*. Eventually, all the compound is converted to the heterocycle.



The mechanism of this reaction is of great importance because it is the foundation stone of the synthesis of Crixivan (indinavir)—an anti-HIV drug that has saved thousands of lives.



➡ The last step looks a long stretch, but you saw in Chapter 31 that 5-*endo-dig* cyclizations are favourable (p. 813).

🔗 Interactive mechanism for the formation of *cis*-aminoindanol by the Ritter reaction

➡ The story of indinavir is recounted in Chapter 43.

Summary of methods for the investigation of mechanism

This brief summary is for guidance only and the figures quoted are approximate ranges only. The full text above should be used for detail. All methods would not be used in one investigation.

1. Make sure of the structure of the product

- Basic structure (Chapters 3, 13, and 18) and stereochemistry (Chapter 31) by spectroscopic methods.
- Detail of the fate of individual atoms by labelling with D, ^{13}C , and ^{18}O . Double labelling may help.
- The stereochemical course of the reaction (enantio- or diastereoselectivity) may be critical.

2. Kinetic methods

- Rate equation gives the composition of main transition state.
- Deuterium isotope effect: $k_{\text{H}} > k_{\text{D}}$ shows bond to H formed and/or broken in transition state. Values of $k_{\text{H}}/k_{\text{D}}$ of 2–7 typical.
- Entropy of activation shows increase ($\Delta S^\ddagger$ positive) or decrease ($\Delta S^\ddagger$ negative) in disorder. Typical values and deductions:
 - $\Delta S^\ddagger$ positive (rarely larger than $+50 \text{ J mol}^{-1} \text{ K}^{-1}$): one molecule breaks into two or three
 - moderate negative values: no change in number of molecules (one goes to one etc.) or bimolecular reaction with solvent

- large negative values: two molecules go to one or unimolecular reaction with ordered transition state (cycloaddition, etc.)

3. Correlation of structure and reactivity

- Replace one group by another of similar size but different electronic demand (CF_3 for CH_3 or OMe for CH_3).
- Systematic Hammett σ/ρ correlation with *m*- and *p*-substituted benzenes:
 - sign of ρ : $+\rho$ indicates electrons flowing into and $-\rho$ electrons flowing out of ring in transition state
- magnitude of ρ shows effect on the benzene ring:
 - large (around 5), charge on ring ($+\rho$, anion; $-\rho$, cation)
 - moderate (around 2–4), charge on atom next to ring—may be gain or loss of conjugation
 - small (<1), ring may be distant from scene of action or ρ may be balance of two ρ s of opposite sign.

4. Catalysis

- pH–rate profile reveals specific acid or base catalysis.
- Rate variation with $[\text{HA}]$ or $[\text{B}]$ at constant pH reveals GAC or GBC.
- Deuterium isotope effect: normal ($k_{\text{H}} > k_{\text{D}}$) shows GA/BC, inverse solvent $k(\text{D}_2\text{O}) > k(\text{H}_2\text{O})$ shows SA/BC.
- GA/BC is termolecular and has large negative entropy of activation.

5. Intermediates

- Independent preparation or, better, isolation from or detection in reaction mixture helps.
- Must show that intermediate gives product under reaction conditions.
- Designed trapping experiments often most convincing.

Further reading

An excellent modern and rather more advanced book is E. V. Anslyn and D. A. Dougherty, *Modern Physical Organic Chemistry*, University Science Books, Sausalito, CA, 2005.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>